Resolution Geometry: Quantum Gravity

A Conditional Derivation of the Quantum State Structure, Causal Localization, and Gravitational Effective Field Theory

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

November 12, 2026 (third edition)

Abstract

A derivation of low-energy quantum gravity is developed from operational records under an explicit hierarchy of state-space, composition, causal-realization, and metric-response assumptions. Operational equivalence first determines which properties and updates descend to the physical state description. Polytope rigidity and an all-dimension composition identity isolate necessary features of a nonclassical locally tomographic theory; a stated reconstruction theorem supplies the complex quantum realization. Positive distinguishability cannot itself produce Lorentzian signature. Causal order and calibrated volume instead determine a metric within an admitted distinguishing Lorentzian realization. A finite-range unitary quantum walk supplies an exact microscopic propagation cone, a discrete operator continuity equation, graded locality, and a controlled free Dirac limit. These constructions establish causal localization, not dynamical gravity by themselves. Signature and dimension are no longer declared. Given a nondegenerate quadratic realization of unspecified signature:

- predictive closure forces Lorentzian signature;
- propagation of the tensor response between separated sources forces at least three space directions;
- stability of bound record carriers forces at most three. This covers classical orbits and also quantum bound states of Gauss-law forces, which the quantum realization itself supplies.

A covariant source functional then identifies the stress response and its Ward identity. Within a local two-derivative symmetric-tensor sector, gauge invariance selects the linearized Einstein operator. With the stated nonlinear consistency assumptions, the leading action is Einstein–Hilbert with universal source coupling. Perturbative quantization yields gravitational effective field theory rather than Einstein-only quantum closure. Relational observables, stress fluctuations, and a branch-phase entanglement prediction connect the construction to measurable records.

The action quantum enters as the anchor converting action to phase, and exchange balance forces one universal value. The thermal spectrum of the massless field then follows, and its shape depends on the dimension. Absolute radiometry of the T^{n+1} law is therefore a quantitative witness of three space dimensions; the differentially calibrated cosmic blackbody spectrum is not. It fixes $\hbar$ only as a measured anchor, not by derivation. Quantitative distinguishability bounds delimit finite-resource inference without identifying small effects with exact observational nullity. The theorem ladder distinguishes results proved here, structural results used from the literature, conditional physical identifications, and unresolved nonperturbative questions.

Two items left open in earlier editions are settled in part. First, existence of the weight over histories is decided for quadratic actions: a degenerate action has a kernel only on its quotient, a definite action admits both a probability and an amplitude, and an indefinite action admits only the amplitude. A Lorentzian action therefore cannot carry a probability weight that is a function of the action; the Euclidean form of linearized gravity is itself

indefinite on the gauge quotient, with signature $(5, 1)$; and conservation of the source is the condition for the free generating functional to exist. Second, the interacting 3+1 limit is typed. For the classical quartic field with smooth well-prepared data the lattice limit is proved at rate a^2 together with two source variations, the quartic interaction in three space dimensions being the endpoint of the energy method. The lattice error is predicted by a computable field $a^2 E_1$ with a cubic-anisotropic quadratic dispersion term, which time-of-flight data convert into a bound on the spacing of that realization. Hidden defect stress and microstructure show that isotropy and well-preparedness are matching data. The quantum scalar limit closes negatively, by the imported triviality theorem, and becomes an exclusion between interaction and resolution, $\lambda(\mu) \leq 1/(b \ln(\Lambda/\mu))$, which the measured scalar coupling satisfies. No unique ultraviolet completion, interacting gauge continuum limit, or anomaly-free quantum constraint representation is asserted.

Contents

1 Operational foundations and the derivation statement **3**

2 Quantum state structure and composition **4**

3 Coherence, reference structure, and distinguishability **5**

4 Causal realization and metric scale **6**

5 A finite-range unitary realization and its free continuum limit **7**

6 Local observable algebras and exact quantum balance **9**

7 Covariant localization and the stress response **9**

8 Controlled limits and source fluctuations **10**

9 The interacting $3+1$ limit: closed class, predictor, and walls **11**
- 9.1 Setting . . . 11
- 9.2 The closed class . . . 11
- 9.3 Predictor and upper bound . . . 12
- 9.4 The class cannot be dropped . . . 13
- 9.5 The quantum scalar sector . . . 14

10 The universal tensor response **15**

11 Existence of the kernel: quotient, positivity, and oscillation **16**

12 Perturbative quantization and anomaly conditions **18**

13 The gravitational effective action **19**

14 Long-range exchange and quantum corrections **20**

15 The action quantum and thermal radiation **20**

16 Relational observables and gravitational constraints **22**

17 Finite-resource inference and ultraviolet nonuniqueness **22**

18 Further obstructions and their domains **24**

19 A coherence-sensitive gravitational prediction **24**

20 The derivation theorem and dependency ladder **25**

21 Status ledger and remaining mathematical questions **27**

22 Conclusion **28**

A Computational verification and reproducibility **29**

B Changes in this edition **30**

C Normalization and approximation conventions **31**

1 Operational foundations and the derivation statement

The object to be derived is a low-energy quantum theory of a dynamical metric coupled to matter, with a controlled relation between physical records, gauge equivalence, causal localization, and observables. A classical metric equation and a quantum state space are not sufficient separately: the source response, composition, and approximation connecting them must also be specified.

The argument is organized by logical dependence rather than by the chronology of a construction. Mathematical statements with proofs are distinguished from structural inputs whose hypotheses are retained. A finite numerical calculation witnesses an identity or an example; it does not establish a reconstruction theorem, a continuum existence theorem, or ultraviolet completeness.

Definition 1.1 (Operational equivalence)**.** Let X be a candidate state space and let P_x^a be the outcome law of a declared protocol $a \in \mathcal{P}$. Define

$$x \sim_{\mathcal{P}} y \quad \Longleftrightarrow \quad P_x^a = P_y^a \quad \text{for every } a \in \mathcal{P}, \qquad Q = X/\!\sim_{\mathcal{P}}. \tag{1}$$

Equality here is exact equality of laws, not failure of a finite experiment to reject a hypothesis.

Theorem 1.2 (Descent of properties and dynamics)**.** *A property $C : X \to Y$ is a property of Q if and only if it is constant on every equivalence class. A deterministic update $U : X \to X$ induces an update on Q if and only if*

$$x \sim_{\mathcal{P}} y \implies Ux \sim_{\mathcal{P}} Uy. \tag{2}$$

Proof. If $C = \bar{C} \circ \pi$ for the quotient map π, equal quotient points have equal C. Conversely, constancy makes $\bar{C}([x]) = C(x)$ well defined. The same argument applied to $\bar{U}([x]) = [Ux]$ proves the dynamical statement. □

For stochastic dynamics, the corresponding condition is equality of pushed-forward transition laws on the quotient. In a finite Markov chain it is strong lumpability: the transition probability into each quotient block is independent of the representative within the starting block. Present agreement alone does not imply this condition.

Assumption A1 (Operational closure)**.** The retained records separate the states relevant to the intended predictions and satisfy predictive descent. If a future experiment distinguishes representatives, the state description is refined rather than assigned an autonomous update prematurely.

Scope of the endpoint. The principal result, Theorem 20.1, is conditional. Its assumptions identify a quantum realization, an admissible Lorentzian continuum, and a universal healthy tensor response. Its conclusion is the gravitational effective field theory (EFT) and its observable low-energy consequences. A microscopic derivation of all these assumptions is a separate existence and selection question.

2 Quantum state structure and composition

Assumption A2 (Finite-dimensional operational systems)**.** Each elementary system under consideration has a finite-dimensional compact convex normalized state space and finite information capacity. Effects are affine outcome-probability functionals; composites and their admissible effects are part of the theory.

Finite capacity and finite dimension are separate requirements. A bound on the number of perfectly distinguishable states does not, without additional hypotheses, imply a finite-dimensional affine state space. Likewise, a finite experimental resolution does not turn every continuum classical model into a simplex. Entropy bounds can motivate finite-capacity descriptions in their domain of applicability, but do not replace these operational assumptions [1].

Assumption A3 (Continuous reversible transitivity)**.** For each elementary closed system, the connected reversible transformation group acts transitively on pure states. The requirement concerns closed-system transformations, not arbitrary reduced dynamics after discarding an environment.

Assumption A4 (Composition and reconstruction conditions)**.** Composite states are determined by joint probabilities of product effects (local tomography). The theory also satisfies the subspace-equivalence and elementary-system measurement conditions of the quantum reconstruction used below. In particular, all effects required by that reconstruction on the elementary bit are allowed. These are substantive assumptions about a family of systems, not consequences of one interference experiment.

Theorem 2.1 (Polytope rigidity)**.** *The affine automorphism group of a finite-dimensional polytope is finite. Every continuous one-parameter group of its affine automorphisms is trivial.*

Proof. Every affine automorphism permutes the finite vertex set. The permutation action is faithful because the vertices affinely span the polytope. The automorphism group therefore embeds in a finite permutation group. A continuous map from a connected parameter interval to that finite group is constant. □

Corollary 2.2 (Exclusion of nontrivial polytopic pure-state motion)**.** *A polytope with more than one pure state cannot satisfy Assumption A3. This excludes finite classical simplices and finite polygon models within the declared class. It does not exclude a classical continuum with unrestricted effects: such a model can have continuous pure-state motion and unbounded capacity.*

Theorem 2.3 (All-dimension matrix-family composition identity)**.** *Within a same-type matrix family with rank d, parameter count*

$$K_k(d) = d + \frac{k}{2}d(d-1), \tag{3}$$

and composite rank $d_{AB} = d_A d_B$, local tomography requires

$$K_k(ab) - K_k(a)K_k(b) = -\frac{k(k-2)}{4}ab(a-1)(b-1) = 0. \tag{4}$$

For $a, b > 1$, the only possibilities are $k = 0$ and $k = 2$.

Proof. Substitution and factorization give (4). The remaining factors are nonzero for nontrivial ranks. □

The identity is exact in all ranks, not a finite-rank extrapolation. The classical count $k = 0$ and the complex matrix count $k = 2$ survive; the real and quaternionic counts $k = 1, 4$ do not satisfy this particular composition rule. The calculation does not classify arbitrary convex theories, arbitrary spin factors, or exceptional Jordan systems. In particular, the exceptional octonionic algebra is not an unrestricted all-rank matrix family.

Structural input 2.4 (Finite-dimensional quantum reconstruction)**.** *The reconstruction of Masanes and Müller uses finite-dimensional operational systems, continuous reversible pure-state transitivity, tomographic locality, equivalence of subspaces, and its stated elementary-system measurement assumption to obtain finite-dimensional complex quantum theory [2]. Assumptions A2–A4 are understood with the full operational content of that theorem, not merely the parameter-count identity. Other reconstructions use different sufficient axiom sets [3, 4].*

Consequently the quantum representation used here is

$$\rho \succeq 0, \quad \operatorname{Tr}\rho = 1, \qquad 0 \preceq E \preceq I, \qquad P(E \mid \rho) = \operatorname{Tr}(\rho E), \qquad \mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \tag{5}$$

This is a conditional identification by a reconstruction theorem. Polytope rigidity and composition counting explain two restrictions but do not substitute for the full reconstruction. The state-space axioms do not determine a gravitational Hamiltonian or select the matter spectrum.

3 Coherence, reference structure, and distinguishability

Fix a reference basis and define dephasing by $\Delta(\rho) = \operatorname{diag}\rho$. Diagonal unitary rephasing acts as $\rho \mapsto D\rho D^\dagger$. The two constructions have different quotients.

Proposition 3.1 (Dephasing fibres and phase orbits)**.** *Dephasing is the uniform diagonal-phase twirl, but equality of dephased states does not imply membership in the same diagonal-unitary orbit.*

Proof. Integrating $e^{i(\theta_i-\theta_j)}\rho_{ij}$ over independent phases kills every off-diagonal entry. However, diagonal rephasing preserves $|\rho_{ij}|$ and the spectrum. The states $(I + aX)/2$ with different $|a|$ all dephase to $I/2$ and have different spectra. They therefore lie in different orbits. □

On a fixed nonzero support graph, diagonal rephasing changes an edge phase by $\theta_i - \theta_j$. A spanning forest removes all tree-edge phases. The remaining independent cycle phases number $|E| - |V| + c$, where c is the number of connected components. Products such as $\rho_{01}\rho_{12}\rho_{20}$ are invariant. Positivity restricts the allowed edge magnitudes and phases; support changes can alter the orbit stratum.

Proposition 3.2 (Operational recovery of off-diagonal structure)**.** *Let $p_i = \rho_{ii}$, and let p^+_{ij} and p^i_{ij} be probabilities of the rank-one effects associated with $(|i\rangle + |j\rangle)/\sqrt{2}$ and $(|i\rangle + i|j\rangle)/\sqrt{2}$. Then*

$$\operatorname{Re}\rho_{ij} = p^+_{ij} - \frac{p_i + p_j}{2}, \qquad \operatorname{Im}\rho_{ij} = \frac{p_i + p_j}{2} - p^i_{ij}. \tag{6}$$

Thus populations and two quadratures for every pair identify the density operator in the declared finite-dimensional realization.

Proof. Expand the two Born probabilities. Hermiticity fixes the conjugate entry. The resulting entries exhaust a Hermitian matrix, subject to trace normalization and positivity. □

Uncertainty in a fixed measurement is not coherence: an eigenstate can have zero uncertainty and zero basis coherence, while a coherent superposition has nonzero uncertainty in that basis. Conversely, a scalar coherence measure does not identify the state or its cycle phases. These distinctions prevent a scalar information density from being used as a complete gravitational source.

On a regular fixed-rank stratum and for tangent directions with finite information, the symmetric logarithmic derivative L satisfies $\dot{\rho} = (\rho L + L\rho)/2$ on the relevant support. The SLD quadratic form is $F_Q = \mathrm{Tr}(\rho L^2)$, with Bures convention $\mathrm{d}s_B^2 = F_Q \mathrm{d}\theta^2/4$ for a one-parameter family. Directional optimization over measurements does not guarantee one measurement jointly attains the full multiparameter matrix.

Proposition 3.3 (A joint single-copy resolution obstruction)**.** *At $\rho = I/2$ with tangent derivatives $X/2, Y/2$, the SLD Fisher matrix is I_2. Every single-copy qubit POVM has classical Fisher matrix satisfying $\mathrm{Tr}\, F_{xy} \le 1$, so cannot attain I_2.*

Proof. Write each nonzero effect as $E_a = w_a(I + n_a \cdot \sigma)$, where $|n_a| \le 1$, $\sum_a w_a = 1$, and $\sum_a w_a n_a = 0$. At the center $p_a = w_a$ and $\partial_i p_a = w_a n_{a,i}$. Hence $F_{xy} = \sum_a w_a n_{a,xy} n_{a,xy}^T$ and $\mathrm{Tr}\, F_{xy} \le \sum_a w_a = 1$. □

The statement is single-copy and local in parameter space. It does not rule out a different collective asymptotic estimation task. Likewise, the Berry curvature of the full pure-state manifold must not be confused with its pullback to a fixed-population fibre. For a pure qubit that fibre is a circle, on which every two-form vanishes; the Chern number belongs to the full projective-state sphere.

4 Causal realization and metric scale

Obstruction 4.1 (Signature cannot arise by a real distinguishability pullback)**.** *If $G \succeq 0$ and J is real, then $J^T G J \succeq 0$. No real pullback of a positive Fisher, SLD, or Bures metric has Lorentzian signature.*

Proof. For every v, $v^T J^T G J v = (Jv)^T G (Jv) \ge 0$. □

A Lorentzian realization therefore requires causal or dynamical structure beyond a positive state-space metric. Analytic continuation or a subtraction rule is additional structure, not a coordinate change proving the desired signature.

Assumption A5 (Admissible causal continuum)**.** The relevant records admit a smooth, time-oriented realization by a nondegenerate quadratic form of unspecified signature (p, q). Field dynamics is second order with that form as principal symbol. A specified correspondence links operational reachability to the causal or chronological relation used in the reconstruction theorem, and a calibrated local volume measure is supplied. Signature and dimension are not assumed: Theorem 4.2 derives them relative to stated premises. Manifoldlikeness and the reachability correspondence are not inferred from an arbitrary partial order alone.

Theorem 4.2 (Signature and dimension)**.** *Under Assumption A5:*

(a) Predictive closure. *A mode $e^{\lambda x_e + ik\cdot x}$ of $\sum_i s_i \partial_i^2 u = 0$ evolving along a direction of sign s_e has $\lambda^2 = s_e \sum_{i \ne e} s_i k_i^2$. Growth is bounded at all frequencies iff $\min(p, q) = 1$. Definite and ultrahyperbolic realizations admit no Lipschitz predictive update.*

(b) Propagation. *Suppose the tensor response transmits exchange between separated sources. On shell it has $D(D-3)/2$ polarizations; the Weyl tensor has no components in $D = 3$; the static exchange numerator $1 - 1/(D-2)$ vanishes there; and $G_{\mu\nu} \equiv 0$ in $D = 2$. Hence $D \ge 4$.*

(c) Stable carriers. *Suppose records are kept by bound carriers of Gauss-law forces, with potential* $\propto r^{-(n-2)}$ *in* $n = D - 1$ *space dimensions. Then:*

- *Newtonian circular orbits are stable only for* $n = 3$*;*
- *Schwarzschild–Tangherlini circular orbits are stable for* $D = 4$ *and for no* $D = 5, 6, 7$*;*
- *in the quantum realization,* $H = -\Delta - g\, r^{-(n-2)}$ *is scale invariant with no ground state at* $n = 4$*, and unbounded below for* $n \geq 5$*.*

Hence $D \leq 4$*.*

The resolved realization is therefore $(1, 3)$*. The statement concerns resolved dimensions; compact directions below every accessible protocol are not excluded.*

Proof. (a) Substitute the mode. If $\min(p, q) \geq 2$, or one of p, q is zero, a frequency direction exists along which $\lambda^2 > 0$ grows without bound, which is Hadamard ill-posedness. (b) Count the kernel of the Einstein symbol at null momentum, subtract the D gauge modes, and use the Weyl component count $D^2(D^2 - 1)/12 - D(D + 1)/2$. (c) Orbits follow from the effective potentials. For the quantum statement, a Gaussian of width s has $E(s) = \frac{n}{2s^2} - g\,\Gamma(1)\Gamma(n/2)^{-1} s^{-(n-2)}$, whose infimum is $-\infty$ for $n \geq 5$. At $n = 4$, dilations map H to $s^{-2}H$, which excludes an isolated negative ground state. These statements are proved and checked as groups X1–X4 of the Tensor Derivation. □

Structural input 4.3 (Causal reconstruction)**.** *Under the distinguishing and regularity hypotheses of the causal reconstruction results, preservation of the appropriate causal/chronological structure determines the Lorentzian conformal geometry [6, 7]. The conclusion applies within an admitted Lorentzian manifold class; it is not an existence theorem assigning a manifold to every ordered set.*

Theorem 4.4 (Calibrated volume fixes the conformal factor)**.** *Suppose two smooth Lorentzian metrics on the same d-manifold have the same time-oriented causal structure under the reconstruction hypotheses and the same local volume measure. Then they coincide under the resulting conformal identification.*

Proof. Causal reconstruction gives $\widetilde{g} = \Omega^2 g$ with $\Omega > 0$. Their volume forms satisfy $\mathrm{d}V_{\widetilde{g}} = \Omega^d \mathrm{d}V_g$. Equality of measures on all relatively compact regions implies $\Omega^d = 1$ almost everywhere, and smoothness gives $\Omega = 1$ everywhere. □

Proposition 4.5 (Statistical volume calibration)**.** *For a Poisson sprinkling of known density* ϱ *into a region of volume* V*,* $\widehat{V} = N/\varrho$ *is unbiased and*

$$\operatorname{Var} \widehat{V} = V/\varrho, \qquad \frac{\operatorname{sd} \widehat{V}}{V} = \frac{1}{\sqrt{\varrho V}}. \tag{7}$$

Proof. The count has mean and variance ϱV. □

Counting supplies a statistical estimate of volume once density and units are calibrated [8]. If ϱ is unknown, the degeneracy $\varrho \mapsto \varrho/\Omega^d$, $g \mapsto \Omega^2 g$ remains. A finite count is not exact geometry. The physical dimension and an interacting continuum limit remain separate obligations.

5 A finite-range unitary realization and its free continuum limit

A local reversible construction establishes compatibility of exact microscopic propagation bounds with a relativistic free limit. It is an existence example, not a unique microscopic law or a four-dimensional gravitational model.

On $\ell^2(\mathbb{Z}) \otimes \mathbb{C}^2$, define

$$C_m = \begin{pmatrix} n & -im \\ -im & n \end{pmatrix}, \quad n = \sqrt{1-m^2}, \quad 0 \le m < 1, \qquad U = SC_m, \tag{8}$$

where S shifts the right component one site right and the left component one site left. This is a coined quantum walk with a Dirac-type low-energy limit. Its precise update is fixed here; distinct Dirac automata can have different finite-spacing dispersion. A low-momentum sector does not remove other Brillouin-zone branches; species multiplicity and fermion doubling require separate analysis in a microscopic matter model.

Theorem 5.1 (Exact finite propagation and dispersion)**.** *The update is unitary and has propagation range one. A state initially supported on A is supported on its t-neighborhood after t integer steps. In Fourier space,*

$$U(k) = e^{-ikZ} C_m, \qquad \cos\omega(k) = n\cos k. \tag{9}$$

For $0 < m < 1$,

$$\left|\frac{\mathrm{d}\omega}{\mathrm{d}k}\right| \le 1, \qquad \omega(k)^2 = \alpha^2 + \frac{\alpha n}{m}k^2 + O(k^4), \quad \alpha = \arcsin m. \tag{10}$$

For $m = 0$, the positive quasienergy branch is $|k|$ on the principal Brillouin interval.

Proof. The coin and conditional shift are unitary. Each step enlarges support by at most one site, so induction proves exact finite propagation. The Fourier matrix has determinant one and trace $2n\cos k$, giving the dispersion. Differentiation gives

$$\omega'(k)^2 = \frac{n^2\sin^2 k}{1 - n^2\cos^2 k} \le 1, \tag{11}$$

because the denominator minus the numerator is m^2. Expanding ω^2 at $k = 0$ gives (10). □

In particular,

$$\alpha^2 = m^2 + \frac{m^4}{3} + O(m^6), \qquad \frac{\alpha n}{m} = 1 - \frac{m^2}{3} - \frac{2m^4}{15} + O(m^6). \tag{12}$$

These are expansions, not exact polynomial dispersion relations. Their even powers do not by themselves certify all observational Lorentz tests for an interacting higher-dimensional model.

Theorem 5.2 (Controlled free Dirac limit on a momentum band)**.** *Let the spatial and temporal spacing be a, set $m = \sin(Ma)$ for fixed $M \ge 0$, and restrict to $|p| \le P$. For $N = T/a \in \mathbb{N}$ and sufficiently small a,*

$$U_a(p) = e^{-iapZ} e^{-iaMX}, \qquad H_D(p) = pZ + MX, \tag{13}$$

satisfy

$$\sup_{|p|\le P} \left\| U_a(p)^N - e^{-iTH_D(p)} \right\| \le TaP|M|. \tag{14}$$

Proof. For bounded self-adjoint A, B, the Duhamel product estimate gives $\|e^{-iaA}e^{-iaB} - e^{-ia(A+B)}\| \le a^2\|[A,B]\|/2$. With $A = pZ$ and $B = MX$, the right side is $a^2|pM|$. A telescoping sum of N products of unitaries gives $Na^2|pM| \le TaP|M|$. Fourier transformation transfers the uniform multiplier bound to the band-limited Hilbert space. □

This closes a free, finite-time, fixed-band convergence statement. It does not control arbitrarily large momentum, interacting renormalization, or a dynamical curved geometry. The state-evolution error in (14) is $O(a)$; an $O(a^2)$ quasienergy correction is a different assertion. Weyl and Dirac automata in higher dimensions provide related constructions under their own graph, isotropy, and representation hypotheses [10, 11].

6 Local observable algebras and exact quantum balance

Let $a_{x,s}$ satisfy the canonical anticommutation relations (CAR). A one-particle unitary induces a CAR automorphism, with field evolution $a(t) = U^t a(0)$ in a consistent convention.

Theorem 6.1 (Graded locality and observable locality)**.** *For the finite-range update,*

$$\{a_{x,s}(t), a^\dagger_{y,r}(0)\} = (U^t)_{xs,yr} = 0 \quad \text{if } |x-y| > t. \tag{15}$$

Odd fields have graded locality. Even observable algebras supported in disjoint causal neighborhoods commute.

Proof. The CAR reduce the unequal-time anticommutator to the one-particle propagator, whose support is bounded by Theorem 5.1. Products of even parity in disjoint CAR supports commute because moving all generators past one another produces an even number of sign changes. □

The even local net is the observable-locality object. The odd field net is graded-local, not an ordinary commuting net. A lattice net with this property is not yet a continuum Haag–Kastler theory with all covariance, spectrum, and representation axioms [9].

Theorem 6.2 (Operator continuity equation)**.** *Let $b_x = C_m a_x$ be the post-coin fields, $n_x = a^\dagger_{x,R} a_{x,R} + a^\dagger_{x,L} a_{x,L}$, and*

$$J_{x+1/2} = b^\dagger_{x,R} b_{x,R} - b^\dagger_{x+1,L} b_{x+1,L}. \tag{16}$$

Then

$$n_x(t+1) - n_x(t) = -\big(J_{x+1/2}(t) - J_{x-1/2}(t)\big) \tag{17}$$

as an operator identity.

Proof. Coin unitarity gives $n_x = b^\dagger_{x,R} b_{x,R} + b^\dagger_{x,L} b_{x,L}$. After the shift, $n_x(t+1) = b^\dagger_{x-1,R} b_{x-1,R} + b^\dagger_{x+1,L} b_{x+1,L}$. Subtraction is exactly (17). □

The identity holds in every state, not only for one simulated wave packet. It is a number-current balance law. Identifying it with energy-momentum conservation requires an additional stress construction; a conserved probability current alone does not identify the gravitational source.

7 Covariant localization and the stress response

A local net assigns observable algebras to already specified regions. It establishes localization and inclusion relations. It does not, by itself, determine a metric, a dynamical tetrad, or the map from microscopic records to a gravitational response.

Assumption A6 (Covariant source realization)**.** In the admitted continuum regime, local matter records admit a covariant renormalized generating functional $W[g, J]$ on admissible backgrounds. The metric argument is the physical metric calibrated by causal and volume records. Its variations exist to the required order, obey the appropriate diffeomorphism Ward identity, and are compatible with the retained microscopic records. Scalar sources J are used below for simplicity. Boundary conditions and additional source transformations are specified when needed.

This assumption is the state-to-spacetime identification needed for a stress response. It is stronger than merely possessing a local net, and is not supplied by an internal phase holonomy.

Theorem 7.1 (Stress response and Ward identity)**.** *Define the matter stress by*

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g^{\mu\nu}}, \tag{18}$$

and the renormalized expectation and scalar response by

$$\delta W = \int \sqrt{-g}\left(\frac{1}{2}\langle T^{\mu\nu}\rangle \delta g_{\mu\nu} + \langle \mathcal{O}\rangle \delta J\right). \tag{19}$$

For compactly supported infinitesimal diffeomorphisms and no diffeomorphism anomaly,

$$\nabla_\mu \langle T^\mu{}_\nu\rangle = \langle \mathcal{O}\rangle \nabla_\nu J. \tag{20}$$

Functional differentiation gives the corresponding stress-insertion Ward identities with contact terms.

Proof. Use $\delta_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$ and $\delta_\xi J = \xi^\nu \nabla_\nu J$. Integrating the metric term by parts in $\delta_\xi W = 0$ gives (20) because ξ is arbitrary. Further source derivatives differentiate both the expectation values and their transformation laws, producing the contact terms. □

Conservation does not fix all source couplings. In flat space,

$$I_{\mu\nu} = (\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu)F, \qquad \partial^\mu I_{\mu\nu} = 0, \tag{21}$$

is an improvement, related to a nonminimal RF coupling in a covariant action. Scalar improvements, spin/torsion-sensitive sectors, and boundary responses must therefore be retained in the source ledger.

Obstruction 7.2 (Internal phase curvature does not determine gravity)**.** *A nonzero cycle phase of a finite-dimensional quantum state does not imply nonzero spacetime Riemann curvature.*

Proof. Take a positive three-level density matrix with nonzero $\arg(\rho_{01}\rho_{12}\rho_{20})$ and realize it as an internal state on a fixed Minkowski background. The internal invariant is nonzero while the background Riemann tensor vanishes. □

The Coleman–Mandula theorem concerns symmetry algebras of an analytic scattering theory under specific hypotheses, including a mass gap [13]. It is not a general proof that a microscopic internal structure can never participate in an emergent spacetime construction, and it is not directly applicable to a massless gravitational theory without checking those hypotheses. The obstruction above is narrower and sufficient: internal holonomy alone does not perform the required identification.

8 Controlled limits and source fluctuations

A continuum limit must control the source and its variations, not only the underlying field values.

Obstruction 8.1 (Nonlinear stress need not commute with a field limit)**.** *Uniform convergence of fields does not imply convergence of their quadratic gradient stresses to the stress of the limiting field.*

Proof. On a periodic interval, $u_n(x) = \sin(nx)/n$ converges uniformly to zero, but

$$\frac{1}{2}|\partial_x u_n|^2 = \frac{1}{2}\cos^2(nx) \rightharpoonup \frac{1}{4}, \tag{22}$$

whereas the limiting field has zero gradient stress. □

A strong enough convergence topology, suitable compactness, or an explicit defect stress is required. At the quantum level, products and stress correlators additionally require renormalization. A finite-range free limit does not settle these interacting source limits; Section 9 settles them for the scalar sector.

The mean-source equation

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle_{\mathrm{ren}} \tag{23}$$

is a semiclassical approximation, not the definition of the full quantum theory. Equal means need not give equal fluctuations. For $H = \mathrm{diag}(0, 1, 2)$, the states $\mathrm{diag}(0, 1, 0)$ and $\mathrm{diag}(1/2, 0, 1/2)$ both have mean energy one but variances zero and one. A quantum source description retains connected stress correlations and their backreaction in the regime where they matter.

9 The interacting $3+1$ limit: closed class, predictor, and walls

The free fixed-band theorem of Section 5 does not reach an interacting field in three space dimensions. This section separates that question into the parts that are mathematics and the parts that are data, and settles the scalar sector. The outcome is typed: a class in which the classical limit is proved with its rate, a computable predictor for the lattice error, obstructions showing that the class cannot be dropped, and a negative closure of the quantum scalar limit that turns into an upper bound.

9.1 Setting

Let $\Lambda_a = (a\mathbb{Z}/\mathbb{Z})^3$ be the periodic cubic lattice of spacing $a = 1/N$, with the mean norms $\|w\|_{p,a}^p = a^3 \sum_x |w(x)|^p$, forward differences ∇_a, and the nearest-neighbour Laplacian Δ_a. The lattice field obeys

$$\ddot{q} = \kappa \Delta_a q - m^2 q - \lambda q^3 + sf, \qquad \kappa, m^2 > 0,\ \lambda \geq 0, \tag{24}$$

with a smooth source f of amplitude s. Its energy and energy norm are

$$E_a = \left\| \tfrac{1}{2}\dot{q}^2 + \tfrac{\kappa}{2}|\nabla_a q|^2 + \tfrac{m^2}{2} q^2 + \tfrac{\lambda}{4} q^4 \right\|_{1,a}, \qquad \|(w, \dot{w})\|_{E,a}^2 = \|\dot{w}\|_{2,a}^2 + \kappa \|\nabla_a w\|_{2,a}^2 + m^2 \|w\|_{2,a}^2. \tag{25}$$

The continuum field u obeys the same equation with Δ in place of Δ_a. The first and second source variations $z = \partial_s q$ and $y = \partial_s^2 q$ obey the linearized equations

$$\ddot{z} = \kappa \Delta_a z - (m^2 + 3\lambda q^2) z + f, \qquad \ddot{y} = \kappa \Delta_a y - (m^2 + 3\lambda q^2) y - 6\lambda q z^2. \tag{26}$$

9.2 The closed class

Lemma 9.1 (Uniform discrete Sobolev bound)**.** *There is a constant C_S, independent of a, such that $\|w\|_{6,a} \leq C_S \big(\|\nabla_a w\|_{2,a}^2 + \|w\|_{2,a}^2 \big)^{1/2}$ for every lattice function w on Λ_a.*

Proof. Let Iw be the piecewise trilinear interpolant. On one cell, the L^6 norm of Iw and the ℓ^6 norm of the eight vertex values are two norms on $\mathbb{R}^8$; the L^2 norm of ∇Iw and the root of the summed squared edge differences are two seminorms on $\mathbb{R}^8$ with the same kernel, the constants. Both pairs are therefore equivalent, and by scaling the constants do not depend on a. Summing over cells gives $\|w\|_{6,a} \leq c \|Iw\|_{L^6}$ and $\|Iw\|_{H^1} \leq c' \|w\|_{H^1_a}$. The Sobolev embedding $H^1(\mathbb{T}^3) \subset L^6(\mathbb{T}^3)$ completes the proof. □

Theorem 9.2 (Interacting refinement in three space dimensions)**.** *Let the continuum data and source be smooth, let u be the smooth solution on $[0,T]$, and let the lattice data be the restriction of the continuum data to Λ_a. Then*

$$\sup_{t\le T} \|(q-u, \dot q-\dot u)\|_{E,a} \le C\, a^2, \tag{27}$$

and the same bound holds for the first and second source variations. The constant depends on T, λ, C_S, the energy, the source, and bounds on fourth space derivatives of the continuum fields, not on a.

Proof. Uniform energy. Since $\dot E_a = s\langle f, \dot q\rangle_a \le s\|f\|_{2,a}\sqrt{2E_a}$ and $\lambda \ge 0$, the energy norm of q is bounded on $[0,T]$ uniformly in a. By Lemma 9.1, so is $\|q\|_{6,a}$.

Consistency. On smooth functions $\Delta_a u = \Delta u + \frac{a^2}{12}\sum_i \partial_i^4 u + O(a^4)$, so the restriction of u solves (24) up to a residual τ with $\|\tau\|_{2,a} \le c\, a^2$.

Stability. The error $e = q - u$ obeys $\ddot e = \kappa\Delta_a e - m^2 e - \lambda(q^3 - u^3) + \tau$. Hölder's inequality with $\frac12 = \frac16 + \frac13$ gives

$$\|q^3 - u^3\|_{2,a} \le \|e\|_{6,a}\, \|q^2 + qu + u^2\|_{3,a} \le \tfrac32 (\|q\|_{6,a}^2 + \|u\|_{6,a}^2)\|e\|_{6,a} \le C' \|(e,\dot e)\|_{E,a}. \tag{28}$$

Hence $\frac{\mathrm d}{\mathrm dt}\|(e,\dot e)\|_{E,a} \le \lambda C' \|(e,\dot e)\|_{E,a} + \|\tau\|_{2,a}$, and Grönwall's inequality with zero initial error gives the bound.

Variations. The error $\zeta = z - Z$ of the first variation is forced by $q^2 z - u^2 Z = q^2\zeta + (q+u)eZ$, and $\|q^2\zeta\|_2 \le \|q\|_6^2\|\zeta\|_6$, $\|(q+u)eZ\|_2 \le \|q+u\|_6\|e\|_6\|Z\|_6$. The second variation adds $qz^2 - uZ^2$, again a sum of products of three factors bounded in L^6, one of which is an error already controlled. The same Grönwall argument applies at each order. □

Global smooth continuum solutions exist for smooth data by Jörgens's theorem [28], which transfers to the torus by finite propagation speed. No smallness of the coupling, the amplitude, or the time is used. Group IB runs $N = 8, 16, 32$ against a spectral reference with $\lambda = 3$ and amplitudes of order two: the state, first-variation, and second-variation errors fall by 3.90, 3.97, and 4.02 on the last halving. Against the continuum law with the interaction removed, the error stalls at the size of the interaction effect. The factors are a check; the proof is the estimate (28).

Remark 9.3 (Endpoint). Estimate (28) uses three factors in L^6, and $H^1 \subset L^6$ holds exactly up to three space dimensions. The quartic interaction in $3+1$ dimensions is therefore the endpoint at which the force is Lipschitz on the energy space. For a higher power, or in four space dimensions, the lattice delta function shows that no uniform bound exists (growth $N^{1/2-3/p}$ for $p > 6$, and $N^{1/3}$ for $p = 6$ in four dimensions; group IA). This is a wall of the method, not a proof of divergence. Convergence for data that are only of finite energy is not claimed.

9.3 Predictor and upper bound

Theorem 9.4 (First lattice correction)**.** *Under the hypotheses of Theorem 9.2, let E_1 solve the linearized continuum equation*

$$\ddot E_1 = \kappa\Delta E_1 - (m^2 + 3\lambda u^2)E_1 + \frac{\kappa}{12}\sum_{i=1}^{3}\partial_i^4 u, \qquad E_1(0) = \dot E_1(0) = 0. \tag{29}$$

Then $q = u + a^2 E_1 + r_a$ with $\sup_{t\le T}\|(r_a, \dot r_a)\|_{E,a} \le Ca^4$. In particular $\|(q-u, \dot q - \dot u)\|_{E,a} \le a^2\|(E_1, \dot E_1)\|_{E,a} + Ca^4$.

Proof. Insert $u + a^2 E_1$ into (24). The terms of order a^2 cancel by the equation for E_1, leaving a residual of order a^4 from the next term of the expansion of Δ_a and from $3\lambda u\,(a^2E_1)^2$. Apply the stability step of Theorem 9.2 to $q - u - a^2E_1$. □

The lattice field is thus described to this order by the modified equation

$$\ddot{u} = \kappa\Delta u + \frac{\kappa a^2}{12}\sum_i \partial_i^4 u - m^2 u - \lambda u^3, \qquad \omega^2 = m^2 + \kappa k^2 - \frac{\kappa a^2}{12}\,k^4 \sum_i n_i^4 + O(a^4), \tag{30}$$

with $n = k/|k|$. This is the first term of the effective lattice action in the sense of Symanzik [29]. The direction factor $\sum_i n_i^4$ takes the values 1, $\frac{1}{2}$, $\frac{1}{3}$ along (100), (110), (111). The first trace of the lattice is therefore a quadratic, subluminal violation of Lorentz invariance with cubic anisotropy. No linear term occurs for this reflection-symmetric realization.

Group IC computes E_1 on the reference. The residual $q - u - a^2 E_1$ is 36%, 8.7%, and 2.3% of the error at $N = 8, 16, 32$, and the error is bounded by $1.1\,a^2\|E_1\|$ on every grid.

Corollary 9.5 (Spacing bound from time-of-flight data)**.** *Let a massless band be realized by the nearest-neighbour cubic stencil. In the convention $E^2 = p^2[1 - (E/E_{\mathrm{QG},2})^2]$, Eq. (30) gives $a = \sqrt{12/\sum_i n_i^4}\,/E_{\mathrm{QG},2}$. The bound $E_{\mathrm{QG},2} > 6 \times 10^{-8} E_{\mathrm{Pl}}$ from GRB 221009A [38] then gives, for the least favourable orientation,*

$$a < 1.0 \times 10^{8}\,\ell_{\mathrm{Pl}} \approx 1.6 \times 10^{-27}\ \mathrm{m}. \tag{31}$$

The bound is specific to the realization. The next-neighbour improved stencil has no a^2 term (group IC), and a general cubic-symmetric local stencil has two coefficients, multiplying $\sum_i \partial_i^4$ and Δ^2. What is forced is the form: even order, with a cubic harmonic. The conversion also depends on the stated dispersion convention by a factor of order one.

9.4 The class cannot be dropped

Obstruction 9.6 (Hidden defect in three dimensions)**.** *The alternating field $q = aA(-1)^{i+j+k}c(t)$ spans an exact invariant subspace of (24) with $s = 0$. Every low Fourier record of it vanishes for all times and $\sup|q| \to 0$, while its exchange energy density tends to $6\kappa A^2$.*

Proof. On the alternating pattern Δ_a acts as $-12/a^2$ and the cube of the pattern is the pattern, so the subspace is invariant and c obeys an ordinary differential equation. Each of the three forward differences equals $\pm 2A$, giving the energy density $\frac{\kappa}{2} \cdot 3 \cdot 4A^2$. □

Obstruction 9.7 (Microstructure fixes the limit cone)**.** *Let the bond stiffness alternate between $\frac{1}{2}$ and $\frac{3}{2}$ on successive lattice planes normal to the third axis. Its mean is 1 in every direction, and the refinement limit of the linear field is $\ddot{u} = \partial_1^2 u + \partial_2^2 u + \frac{3}{4}\partial_3^2 u$. A refinement sequence alternating between this lattice and the uniform one has two limits.*

Proof. Along the planes the stiffnesses add in parallel, giving the arithmetic mean. Across the planes they add in series, giving the harmonic mean $2/(2 + \frac{2}{3}) = \frac{3}{4}$. Group ID solves the cell problem and the Bloch long-wave problem directly. □

These are defect measures and homogenization limits [30, 31, 32], identified and not new. Their consequence for the present construction is specific. Isotropy of the limiting cone, and the absence of defect stress, are not consequences of refinement. They are matching data, to be declared with the realization, in the same way as the calibrated volume.

Proposition 9.8 (Limit criterion)**.** *Let $\Gamma(r)$ be the set of cluster points of a record r along a refinement family. The record has a limit exactly when $\Gamma(r)$ is one point. A linear target L is determined in the limit by retained linear records M exactly when $\ker M \subseteq \ker L$.*

Proof. The first statement is the definition of convergence in a compact record range. For the second, L factors through M exactly when it vanishes on $\ker M$. □

This is the descent criterion (2) applied to the refinement tail. Obstruction 9.6 is a failure of the second statement: the zero field and the defect field have the same low records and different energies. Obstruction 9.7 is a failure of the first.

9.5 The quantum scalar sector

Obstruction 9.9 (The vacuum is not well prepared)**.** *In the ground state of the free lattice field each mode carries* $\frac{1}{2}\hbar\omega_k$*. Then* $\langle q^2\rangle = \sum_k \hbar/(2\omega_k)$ *grows like* a^{-2}*, the energy density like* a^{-4}*, and the leading interaction shift of the mass,* $3\lambda\langle q^2\rangle$*, diverges. No quantum state satisfies the uniform energy hypothesis of Theorem 9.2.*

Proof. Mode counting on the lattice: the sums are dominated by $|k| \sim 1/a$, where $\omega_k \sim \sqrt{\kappa}/a$ and the number of modes per unit volume is a^{-3}. Group IE evaluates both sums; the ratios per halving are 3.9993 and 15.9997. Every state has at least the ground-state energy. Subtracting that energy does not remove the fluctuation $\langle q^2\rangle$, which is what enters the interaction. □

The classical closure therefore does not lift. The parameters must be renormalized, and the question becomes whether an interacting limit survives.

Structural input 9.10 (Marginal triviality)**.** *For the one-component* $\lambda\phi^4$ *field with a lattice cutoff in four dimensions, and for nearest-neighbour ferromagnetic Ising-type models, every scaling limit at or near the critical point is Gaussian [33].*

Proposition 9.11 (Interaction–resolution bound)**.** *Let the quartic coupling run as* $\mathrm{d}\lambda/\mathrm{d}\ln\mu = b\lambda^2$ *with* $b > 0$ *between a resolved scale* μ *and a cutoff* Λ*. Then for every bare coupling* $\lambda(\Lambda) \in (0, \infty]$*,*

$$\lambda(\mu) \le \frac{1}{b\ln(\Lambda/\mu)}, \qquad \textit{equivalently} \qquad \Lambda \le \mu \exp\Big(\frac{1}{b\,\lambda(\mu)}\Big). \tag{32}$$

Proof. Integration gives $1/\lambda(\mu) = 1/\lambda(\Lambda) + b\ln(\Lambda/\mu)$, and the first term is nonnegative. □

The one-loop flow is a predictor; the rigorous statement that the interaction vanishes with the cutoff is Structural input 9.10, and nonperturbative lattice bounds are available [34]. Read in the present terms, Eq. (32) is an exclusion between two quantities: in the quartic scalar sector, a nonzero interaction at resolved scales and unlimited resolution are incompatible. Finite resolution is forced there, with a computable ceiling.

Corollary 9.12 (Measured scalar coupling)**.** *For* $V = \lambda(H^\dagger H)^2$ *the scalar-sector coefficient is* $b = 3/(2\pi^2)$*. The measured* $m_H = 125.20 \pm 0.11$ *GeV [37] and* $v = 246.22$ *GeV give* $\lambda = m_H^2/(2v^2) = 0.1293$*. This lies below the bound (32) for a Planck-scale cutoff,* 0.168*, which corresponds to* $m_H < 143$ *GeV. The scalar sector alone admits resolution up to* 10^{24} *GeV.*

Bounds of this kind were stated before the measurement [35]; with the full Standard-Model flow at two loops the Planck-cutoff bound is $m_H < 180 \pm 4 \pm 5$ GeV [36]. The number 143 GeV uses the scalar self-coupling alone. It is a consistency witness for the mechanism, not a statement about the full Standard-Model flow, in which the top coupling acts with the opposite sign. Triviality is proved for one component; for the four-component field it is identified, from perturbation theory and lattice computation, and not proved.

Remark 9.13 (What the sign decides). For $b < 0$ Proposition 9.11 has no content: every resolved coupling is reached from a finite bare coupling for every cutoff (group IE). Asymptotically free gauge and fermion sectors are therefore not closed by this argument in either direction.

Sector of the $3+1$ limit	Status	Content
Classical quartic field, smooth well-prepared data, uniform lattice	PROVED	Rate a^2 for the state and two source variations (Theorem 9.2)

Sector of the $3+1$ limit	Status	Content
Lattice error	PREDICTED	a^2E_1; cubic-anisotropic quadratic dispersion; spacing bound (Theorem 9.4, Corollary 9.5)
Outside the class	OBSTRUCTED	Hidden defect stress; anisotropic cone; no full-sequence limit
Higher power, four space dimensions, finite-energy data	METHOD WALL	Energy-space Lipschitz bound fails; not decided
Quantum one-component scalar	CLOSED, NEGATIVE	Limit is Gaussian (imported); bound (32)
Quantum four-component scalar	IDENTIFIED	Same mechanism; not proved
Quantum gauge and fermion sectors	OPEN	Asymptotically free; no bound from this argument

10 The universal tensor response

Assumption A7 (Leading gravitational sector)**.** In the Lorentzian four-dimensional regime of Theorem 4.2, the long-range gravitational response is carried by one healthy massless symmetric tensor. Its leading local parity-even kinetic operator has two derivatives, the matter response is that of Assumption A6, and gauge relabelings preserve physical coincidences. Additional fields and nonminimal couplings are declared explicitly. Healthiness and nonlinear consistency remain assumptions. Within the local first-derivative class, the field content is the single symmetric compensator of local relabeling (Tensor Derivation). The dimension comes from Theorem 4.2, not from positive distinguishability.

Theorem 10.1 (Linear tensor selection)**.** *Let $v_\nu = \partial^\alpha h_{\alpha\nu}$ and $h = \eta^{\mu\nu}h_{\mu\nu}$. In the class*

$$\begin{aligned}\mathcal{E}_{\mu\nu}(h) = {} & A\Box h_{\mu\nu} + B(\partial_\mu v_\nu + \partial_\nu v_\mu) + C\partial_\mu\partial_\nu h \\ & + \eta_{\mu\nu}(D\partial\cdot v + E\Box h),\end{aligned} \tag{33}$$

gauge invariance under $\delta h_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$ and the identity $\partial^\mu\mathcal{E}_{\mu\nu} = 0$ select

$$(A, B, C, D, E) = A(1, -1, 1, 1, -1). \tag{34}$$

The resulting operator is proportional to the linearized Einstein tensor.

Proof. The divergence has independent coefficients $A+B$, $B+D$, and $C+E$, so $B=-A$, $D=A$, and $E=-C$. On a pure gauge field, the coefficients of $\partial_\mu\partial_\nu\partial\cdot\xi$ and $\eta_{\mu\nu}\Box\partial\cdot\xi$ are respectively $2(B+C)$ and $2(D+E)$. Their vanishing gives $C=A$, completing the selection. Substitution verifies both identities for arbitrary h and ξ. □

The overall healthy sign and normalization remain to be fixed. In the transverse-traceless wave sector in four dimensions, the spatial symmetric tensor has six components; transversality to a fixed wave vector and tracelessness leave two independent polarizations. In D dimensions the count is $D(D-3)/2$. This is a linear physical-sector count, not a proof of nonlinear constraint closure.

Proposition 10.2 (Conserved-source gauge decoupling)**.** *The linear coupling $\frac{\kappa}{2}\int h_{\mu\nu}T^{\mu\nu}$ is invariant under compactly supported linearized gauge transformations when $\partial_\mu T^{\mu\nu} = 0$.*

Proof. Its variation is $\kappa\int\partial_\mu\xi_\nu T^{\mu\nu} = -\kappa\int\xi_\nu\partial_\mu T^{\mu\nu}$. □

Structural input 10.3 (Universal coupling and nonlinear completion)**.** *For a local healthy massless spin-two field with the declared field content and derivative order, consistent self-coupling and diffeomorphism invariance give the Einstein leading sector, up to the cosmological term, boundary terms, field redefinitions, and declared nonminimal matter couplings. A canonical uniqueness route uses the hypersurface-deformation algebra with its locality, phase-space, and regularity hypotheses [19]. The full nonlinear consistency result is used as structural input; the linear proof above is not presented as its replacement.*

In an admissible Lorentz-invariant factorizing scattering regime, the leading soft spin-two factor has the form

$$S_{\text{soft}} \propto \sum_i \eta_i g_i \frac{p_i^\mu p_i^\nu \epsilon_{\mu\nu}}{p_i \cdot q}. \tag{35}$$

Gauge decoupling requires $\sum_i \eta_i g_i p_i^\nu = 0$. For generic connected processes, comparison with momentum conservation requires common coupling. This is a soft-factor argument under scattering assumptions, not a global scattering construction for arbitrary cosmological backgrounds.

Corollary 10.4 (Classical leading equation)**.** *Under Assumptions A6 and A7 and Structural input 10.3, choose*

$$S_{\text{EH}} = \frac{2}{\kappa^2} \int \sqrt{-g}(R - 2\Lambda), \qquad \kappa^2 = 32\pi G. \tag{36}$$

Stationarity of $S_{\text{EH}} + S_m$, with the required boundary treatment, gives

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{37}$$

Proof. Variation with respect to $g^{\mu\nu}$ gives $2(G_{\mu\nu} + \Lambda g_{\mu\nu})/\kappa^2 - T_{\mu\nu}/2 = 0$. The stated normalization follows. □

The Bianchi identity supplies compatibility with covariant conservation. It does not determine the numerical values of G, Λ, or the nonminimal couplings.

11 Existence of the kernel: quotient, positivity, and oscillation

Assumption A6 asks that a generating functional exist together with its variations. For a quadratic action this is a finite question with a complete answer, and the answer ties three earlier items together: the gauge quotient, the conserved source, and the signature of Theorem 4.2.

Definition 11.1 (Kernel of an action)**.** Let S be a real function on a finite-dimensional space of histories V. A *kernel* is a weight $w(S(x))$ depending on the history through the action alone, with one action scale, whose total $\int_V w(S)\,\mathrm{d}x$ is finite and nonzero, either absolutely or as the limit $\varepsilon \to 0$ of $\int w(S)\, e^{-\varepsilon x^\top B x/2}\mathrm{d}x$ for positive B. It is a *probability kernel* if $w \geq 0$, and the *amplitude kernel* if $w(S) = e^{iS/\hbar}$.

Lemma 11.2 (Existence of quadratic kernels)**.** *Let $S(x) = \frac{1}{2}x^\top A x$ on $V = \mathbb{R}^n$ with A symmetric, and set $\hbar = 1$.*

(a) If $\ker A \neq 0$, no kernel exists on V. Kernels exist only on the quotient $V/\ker A$.

(b) If A is positive definite, the probability kernel e^{-S} and the amplitude kernel both exist. Their totals are $(2\pi)^{n/2}(\det A)^{-1/2}$ and the same number times $e^{i\pi n/4}$.

(c) If A is nondegenerate of signature (p, q) with $p, q \geq 1$, no probability kernel exists: for every measurable $w \geq 0$ the total is 0 or ∞. The amplitude kernel exists, with

$$\lim_{\varepsilon \to 0} \int e^{iS - \varepsilon x^\top B x/2}\mathrm{d}x = \frac{(2\pi)^{n/2}}{|\det A|^{1/2}}\, e^{i\pi(p-q)/4}, \tag{38}$$

independently of the positive regulator B, and second moments $\langle x x^\top \rangle = iA^{-1}$.

Proof. (a) The integrand is constant along $\ker A$, so the absolute total is 0 or ∞; the regulated total carries a factor $(2\pi/\varepsilon)^{1/2}$ for each null direction and has no limit.

(b) and the second half of (c). Diagonalize A relative to B; by Sylvester's law the relative eigenvalues μ_j have the signature of A. The regulated integral is $(\det B)^{-1/2}\prod_j(2\pi/(\varepsilon-i\mu_j))^{1/2}$ with the principal root, and $(\varepsilon-i\mu)^{-1/2}\to|\mu|^{-1/2}e^{\pm i\pi/4}$ for $\mu\gtrless 0$. Since $\det B\prod|\mu_j|=|\det A|$, the limit is (38). Differentiating in a linear source gives the moments.

First half of (c). Write $2S=|u|^2-|v|^2$ with $u\in\mathbb{R}^p$, $v\in\mathbb{R}^q$. For fixed v, the set $\{s<S<s+\delta\}$ is a shell in u of volume at least $c\,\delta\,|v|^{p-2}$ for large $|v|$. Integrating over v gives $\int^\infty r^{p+q-3}\mathrm{d}r=\infty$ since $p+q\geq 2$. Every level shell of S has infinite measure, so a nonnegative function of S that is positive on a set of positive measure has infinite total. □

Group KA checks each clause: the logarithmic growth of the level shells for signature $(1,1)$ ($4\delta\ln 10$ per decade of radius), the limit (38) for five signatures and for a second regulator, the $\varepsilon^{-1/2}$ divergence along a null direction, and the complex covariance.

Theorem 11.3 (An indefinite action admits only the amplitude)**.** *Suppose that histories are weighted by a kernel of the action, and that the quadratic part of the action is nondegenerate and indefinite on the histories that are integrated over. Then the kernel is not a probability kernel. The amplitude kernel exists, and its second moments are imaginary multiples of the inverse form.*

Proof. Lemma 11.2(c). □

The action of a field in the Lorentzian realization of Theorem 4.2 is of this kind: time and space gradients enter with opposite signs. On a lattice of 16×4 points the free form has 54 positive and 10 negative directions, while its Euclidean continuation is positive (group KB). The statement is conditional and its scope is narrow. It does not derive the Born rule or the reconstruction conditions of Structural input 2.4; it shows that, once a weight over histories is a function of a Lorentzian action, it cannot be a probability and must be an amplitude. Predictive closure, which forces the signature, thereby also excludes a classical probability weight that depends on the history through the action alone. In the definite case both kernels exist, and they differ by the phase $e^{i\pi n/4}$ of clause (b), which is the finite-dimensional content of the Euclidean continuation.

Proposition 11.4 (The gravitational field has no probability kernel)**.** *In D Euclidean dimensions, at momentum $p\neq 0$, the quadratic form of the operator of Theorem 10.1,*

$$Q(h)=\tfrac{1}{2}p^2h_{\mu\nu}h^{\mu\nu}-p^\mu h_{\mu\nu}p_\rho h^{\rho\nu}+p^\mu p^\nu h_{\mu\nu}\,h-\tfrac{1}{2}p^2h^2, \tag{39}$$

has D null directions $p_\mu\xi_\nu+p_\nu\xi_\mu$, exactly one negative direction, and $\frac{1}{2}D(D+1)-D-1$ positive ones. The negative direction is $\theta_{\mu\nu}=\delta_{\mu\nu}-p_\mu p_\nu/p^2$, with $Q(\theta)=-\frac{1}{2}(D-1)(D-2)p^2$. For $D=4$ the signature on the gauge quotient is $(5,1)$.

Proof. Gauge invariance gives the null directions. On the transverse complement, $Q(h)=\frac{1}{2}p^2(h_{\mu\nu}h^{\mu\nu}-h^2)$, which is positive on the traceless part and equals $\frac{1}{2}p^2\big((D-1)-(D-1)^2\big)$ on θ; the two parts are orthogonal for Q. □

By Lemma 11.2(a) the gravitational kernel lives on the gauge quotient, and by clause (c) it is not a probability kernel there, even after Euclidean continuation. This is the conformal-factor problem [27], identified. The amplitude kernel exists on the quotient at this order. In contrast, the Euclidean lattice action of the quartic scalar has a positive Hessian at every field, so its probability kernel exists; this is why the scalar sector could be settled in Section 9 by methods of probability, and why those methods do not extend to the tensor field.

Proposition 11.5 (Conservation is the existence condition)**.** *Add a linear source $J^\top x$ to a quadratic action with* $\ker A \neq 0$*. The normalized generating functional has a nontrivial limit exactly when J annihilates* $\ker A$*, and then $W[J] = -\frac{1}{2} J^\top A^+ J$ on the quotient. For the tensor field this condition is $p_\mu T^{\mu\nu} = 0$.*

Proof. The regulated ratio is $\exp(-\frac{1}{2} J^\top (\varepsilon - iA)^{-1} J)$. A component J_0 of J along $\ker A$ contributes $\exp(-J_0^2/2\varepsilon) \to 0$; otherwise the limit is $\exp(-\frac{i}{2} J^\top A^+ J)$. The pairing of T with $p_\mu \xi_\nu + p_\nu \xi_\mu$ is $2\xi_\nu p_\mu T^{\mu\nu}$. □

The Ward identity (20) is thus not only a consequence of Assumption A6. At quadratic order it is the condition for the functional to exist, and it is the descent criterion (2) for the source: the source must factor through the gauge quotient. Group KB confirms that the regulated response converges for a conserved source and grows like $1/\varepsilon$ otherwise. Assumption A6 is thereby discharged at the free level and remains an assumption for interacting sources.

12 Perturbative quantization and anomaly conditions

Assumption A8 (Perturbative quantum consistency)**.** The admitted quantum theory is quantized around a suitable background with a controlled low-energy expansion, gauge fixing, ghosts, renormalization conditions, and no uncanceled gauge or diffeomorphism anomaly in the specified matter content. Predictions are defined through gauge-independent observables with an appropriate infrared prescription.

Write $g = \bar{g} + \kappa h$. The formal generating functional is

$$Z[J] = \int \mathcal{D}h\, \mathcal{D}\psi\, \mathcal{D}\bar{c}\, \mathcal{D}c\; e^{i(S_{\mathrm{EFT}} + S_{\mathrm{gf}} + S_{\mathrm{gh}} + J\cdot\mathcal{O})}. \tag{40}$$

This denotes a perturbative construction. It is not a convergent nonperturbative Lorentzian measure. BRST/Ward identities, physical-state positivity, and regulator dependence must be controlled at the order used.

Proposition 12.1 (Anomaly controls for one Standard-Model generation)**.** *For left-handed Weyl multiplets Q, u^c, d^c, L, e^c with hypercharges $1/6, -2/3, 1/3, -1/2, 1$, respectively, the $U(1)_Y$ mixed gauge–gravitational, cubic hypercharge, $SU(2)^2U(1)$, and $SU(3)^2U(1)$ anomaly sums vanish. The $SU(3)^3$ chiral anomaly cancels, and the number of $SU(2)$ doublets is even.*

Proof. Including color and weak multiplicities,

$$\operatorname{Tr} Y = 6/6 - 3(2/3) + 3/3 - 2/2 + 1 = 0, \tag{41}$$

$$\operatorname{Tr} Y^3 = 6(1/6)^3 + 3(-2/3)^3 + 3(1/3)^3 + 2(-1/2)^3 + 1 = 0. \tag{42}$$

Up to common Dynkin-index factors, the mixed sums are $3(1/6) - 1/2 = 0$ and $2(1/6) - 2/3 + 1/3 = 0$. The cubic color count is $2 - 1 - 1 = 0$, and the weak-doublet count is $3 + 1 = 4$. Removing e^c makes both hypercharge traces equal to -1. □

These are specified matter-sector checks. The first is a mixed $U(1)$–gravity anomaly, not a proof of all gravitational consistency. The familiar four-dimensional local pure diffeomorphism anomaly is not the same anomaly polynomial. Global gauge-group choices, boundaries, additional matter, and nonperturbative gravitational constraints require their own treatment. The matter spectrum is an empirical input, not selected by the cancellation arithmetic alone.

13 The gravitational effective action

Proposition 13.1 (Derivative power counting)**.** *For a connected graph in four dimensions with L loops and vertices carrying d_v derivatives, the superficial momentum degree is*

$$D = 2 + 2L + \sum_v (d_v - 2). \tag{43}$$

Proof. With I internal lines and V vertices, $D = 4L - 2I + \sum_v d_v$. Use $I = L + V - 1$. □

Power counting identifies allowed counterterms, not their nonzero coefficients. For pure Einstein gravity with vanishing cosmological constant, the one-loop on-shell ultraviolet divergence cancels modulo the four-dimensional Euler density and field redefinitions [14]. This is not a statement that every one-loop amplitude is finite: infrared divergences, matter, boundaries, and different backgrounds require separate analysis.

Structural input 13.2 (Two-loop Einstein nonclosure)**.** *Pure four-dimensional Einstein gravity has a nonzero two-loop on-shell ultraviolet counterterm cubic in the Riemann tensor [15]. Its precise coefficient depends on action and regulator conventions; the nonzero result is the structural fact used here.*

For the curvature convention of the accompanying symbolic calculation, Schwarzschild gives

$$R = 0, \quad R_{\mu\nu}R^{\mu\nu} = 0, \quad R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48M^2}{r^6}, \quad R^{ab}{}_{cd}R^{cd}{}_{ef}R^{ef}{}_{ab} = \frac{96M^3}{r^9}. \tag{44}$$

The cubic invariant is nonzero on a Ricci-flat background. Its value is a witness, not by itself a proof of a loop coefficient or a complete cohomological nonremovability classification; that conclusion uses Structural input 13.2.

Theorem 13.3 (Conditional low-energy quantum gravitational action)**.** *Under Assumption A8, the symmetry-compatible low-energy action is an EFT of the form*

$$S_{\mathrm{EFT}} = \int \sqrt{-g}\left[\frac{M_{\mathrm{Pl}}^2}{2}(R - 2\Lambda) + \mathcal{L}_m + \sum_i c_i \mathcal{O}_i^{(4)} + \sum_j \frac{d_j}{M^2}\mathcal{O}_j^{(6)} + \cdots\right], \qquad M_{\mathrm{Pl}}^{-2} = 8\pi G. \tag{45}$$

For a fixed field content, specified observable, and fixed EFT power-counting order, finitely many independent coefficients enter. The power counting includes canonical dimension or a specified field-amplitude expansion as well as derivatives and loops; derivative order alone need not bound arbitrary scalar-field functions. Einstein-only perturbative closure is excluded by Structural input 13.2; low-energy predictivity is not excluded [16].

Proof. Local counterterms and matching terms are organized by the symmetries and derivative order. At a fixed order in that power counting the independent operator basis is finite after identities, integrations by parts, and field redefinitions. Renormalization fixes scale dependence; matching or measurements fix the required finite coefficients. The nonzero cubic counterterm cannot be absorbed by renormalizing only G and Λ. □

In four dimensions the curvature-squared coefficients are dimensionless. The Euler density, total derivatives, and field redefinitions reduce the independent basis; with matter, the induced contact and nonminimal terms must be retained. Light loops also generate nonlocal terms such as curvature times $\log(-\Box/\mu^2)$ times curvature. Their coefficients depend on the light spectrum and observable.

The validity window requires small external momenta, curvature scales, and gradients relative to the cutoff, together with a controlled loop expansion. Near a strong-curvature region, an EFT prediction requires a new error estimate rather than formal continuation of a truncated series.

Obstruction 13.4 (A derivative truncation is not an exact ultraviolet propagator)**.** *The exact toy inverse operator gives*

$$\frac{1}{x(1+x/m^2)} = \frac{1}{x} - \frac{1}{x+m^2}, \tag{46}$$

with opposite pole residues. Its low-energy expansion is instead

$$\frac{1}{x} - \frac{1}{m^2} + \frac{x}{m^4} - \cdots . \tag{47}$$

An extra pole inferred by exact resummation of a truncated EFT is not automatically a physical state within the EFT window.

A fundamental nondegenerate higher-time-derivative theory and a perturbative EFT have different consistency questions. The spin-two ghost in the usual exact quadratic-gravity treatment is a genuine warning [21], but it does not exclude every curvature correction. For example, $R + \alpha R^2$ with $\alpha > 0$ and positive conformal factor has a healthy local scalar representation with $m_s^2 = 1/(6\alpha)$ in $M_{\rm Pl} = 1$ units. Degenerate constrained theories also require separate analysis.

14 Long-range exchange and quantum corrections

Proposition 14.1 (Newton normalization)**.** *For the canonically normalized massless tensor with $\kappa^2 = 32\pi G$, the four-dimensional conserved-source numerator is*

$$P_{\mu\nu,\alpha\beta} = \tfrac{1}{2}(\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta}). \tag{48}$$

Two static sources give

$$\widetilde{V}(\mathbf{q}) = -\frac{4\pi G m_A m_B}{\mathbf{q}^2}, \qquad V(r) = -\frac{G m_A m_B}{r}. \tag{49}$$

Proof. The numerator contraction is $m_A m_B/2$. The two $\kappa/2$ vertices give $-\kappa^2 m_A m_B/(8\mathbf{q}^2)$. Fourier transformation uses $1/\mathbf{q}^2 \mapsto 1/(4\pi r)$. Conserved sources remove longitudinal gauge terms. □

For minimally coupled massive sources in the conventional nonrelativistic scattering-potential matching prescription, the familiar long-range expression is [17]

$$V(r) = -\frac{G m_1 m_2}{r}\left[1 + 3\frac{G(m_1+m_2)}{rc^2} + \frac{41}{10\pi}\frac{G\hbar}{r^2c^3} + \cdots\right]. \tag{50}$$

The potential beyond leading order is a matching object, not a generally covariant local observable. Coordinate/canonical conventions and subtraction of iterated Born terms matter. The robust EFT content is the nonanalytic long-distance amplitude and its properly defined observable consequences. The quoted coefficient is not asserted for every matter spectrum or every potential convention.

With the stated convention the relative quantum term is $41\ell_P^2/(10\pi r^2)$, where $\ell_P^2 = G\hbar/c^3$. It is approximately 3.41×10^{-70} at one meter and 1.26×10^{-61} at $52\,\mu$m. These are calculated scales, not measured deviations. Local Wilson coefficients remain independent matching data even when a particular long-distance nonanalytic contribution is fixed.

15 The action quantum and thermal radiation

Definition 15.1 (Action–phase anchor)**.** In the complex realization, a continuous one-parameter group of closed-system transformations is $U(t) = e^{-iHt/\hbar}$. Here H is measured in mechanical energy units through exchange records, and $\hbar$ converts action into phase. The reconstruction theorem fixes the existence of the phase group but not the value of $\hbar$.

Proposition 15.2 (Universality of the action quantum)**.** *Let two interacting sectors have action quanta $\hbar_A$ and $\hbar_B$. A resonant exchange of one quantum at common phase frequency ω removes $\hbar_A\omega$ from A and adds $\hbar_B\omega$ to B. Energy balance in the exchange records therefore requires $\hbar_A = \hbar_B$.*

Proof. Resonance means equal phase frequencies, so the time-translation generator is conserved. The mechanical energy ledger changes by $(\hbar_B - \hbar_A)\omega$ per exchanged quantum. In the battery toy the relative drift is 0, 10^{-3} and 10^{-2} for ratios 1, 1.001 and 1.01. □

This parallels universality of the gravitational coupling: one record ledger forces one constant. The value of $\hbar$ remains an anchor; since 2019 it is fixed by the SI definition and realized by measurement.

Theorem 15.3 (Thermal radiation in n space dimensions)**.** *Consider a massless vector field with $n-1$ polarizations in a Gibbs state at temperature T, with Bose statistics and box quantization in the large-volume limit. Its spectral energy density is*

$$u_n(\nu, T) = (n-1)\, S_{n-1}\, \frac{h\nu^n}{c^n}\, \frac{1}{e^{h\nu/k_BT} - 1}, \qquad S_{n-1} = \frac{2\pi^{n/2}}{\Gamma(n/2)}. \tag{51}$$

- *The frequency peak $x = h\nu/k_BT$ solves $x = n(1 - e^{-x})$.*
- *The wavelength peak $x = hc/\lambda k_BT$ solves $x = (n+2)(1 - e^{-x})$.*
- *The total energy density scales as T^{n+1}.*

For $n = 3$ the prefactor is 8π, $x_\nu = 2.821439$, $x_\lambda = 4.965114$, and the Wien constant is $b = hc/(x_\lambda k_B) = 2.897771955 \times 10^{-3}\, m\,K$ from the exact SI constants. For $n = 2, 4, 5$ the Wien constant would be 3.670, 2.404 and $2.057 \times 10^{-3}\, m\,K$.

Proof. Lattice wavevectors in an n-box number $\propto K^n$, so the mode density is $\propto \nu^{n-1}$, with the surface factor S_{n-1} and $n-1$ transverse polarizations. The Bose mean energy per mode is $h\nu/(e^{h\nu/k_BT} - 1)$. Differentiating $\nu^n/(e^x - 1)$, or its wavelength form, gives the two peak equations. Integration gives the T^{n+1} law. The battery checks the mode-count exponents 2.00, 3.00 and 4.00, the roots, and b. □

Corollary 15.4 (A dimension witness from absolute radiometry)**.** *By Theorem 15.3 the radiant exitance of a cavity scales as T^{n+1}, and for $n = 3$ its coefficient is $\sigma = 2\pi^5 k_B^4/(15h^3c^2) = 5.670374 \times 10^{-8}\, W\,m^{-2}\,K^{-4}$ from the exact SI constants. Absolute cryogenic radiometry of a cavity between 233 and 373 K gave $\sigma = (5.66967 \pm 0.00076) \times 10^{-8}\, W\,m^{-2}\,K^{-4}$ [25], which agrees at 0.93σ. Here h and k_B are realized by the Kibble balance and acoustic gas thermometry, not by radiometry, so the comparison is independent. If the exitance scaled as $T^{4+\delta}$, a T^4 inversion would offset the radiometric temperature by $(T/T_0)^{\delta/4}$. Agreement with gas thermometry to $10\,mK$ at $373\,K$ bounds $|\delta| < 3.4 \times 10^{-4}$. The exponent $n + 1 = 4$ forces $n = 3$, and this agrees independently with Theorem 4.2. Gravitational witnesses of $D = 4$ that do not involve electromagnetism are collected in the companion print* Observer Bounds.

Remark 15.5 (Correction: the cosmic spectrum is not a dimension witness). The first printing of this edition used the COBE/FIRAS spectrum as a dimension witness. That inference is withdrawn as circular. FIRAS measures the sky minus an external blackbody calibrator whose spectrum is computed with Planck's law at the thermometer temperature. When the sky is a blackbody at the calibrator temperature, the output vanishes for every n and every instrument response; the battery canary checks this. The FIRAS result ($T = 2.728 \pm 0.004$ K; rms deviation below 50 ppm of the peak; $|y| < 15 \times 10^{-6}$; $|\mu| < 9 \times 10^{-5}$ [26]) therefore certifies that the cosmic background is cavity radiation. It does not certify the law of cavity radiation. The synthetic two-parameter fit in the battery ($n = 3$ residual 4.9×10^{-5} of the peak, best $n = 4$ residual 6.2×10^{-2}) is a property of the two functional forms only.

Remark 15.6 (What the spectrum does not do). The spectrum does not derive $\hbar$. A measured Wien constant gives $h = x_\lambda k_B b/c$ only once c and k_B are fixed. A "peak photon energy" computed as $h\nu$ cannot then be used to infer h without circularity. Moreover, c/λ_{peak} is not the frequency peak: the two differ by the factor 1.76. The thermal graviton spectrum has the same form with $D(D-3)/2 = 2$ polarizations; no measurable thermal graviton background is claimed. With $\hbar$, G and c anchored, $\ell_P = \sqrt{\hbar G/c^3}$ sets the scale of the illustrative bounds in Section 17.

16 Relational observables and gravitational constraints

Gauge-invariant predictions are functions of physical records, not coordinate labels. In canonical language, physical observables commute with first-class constraints on the physical state space, or are represented by an appropriate relational/dressed construction.

Proposition 16.1 (Relational exchange example)**.** *For $C = Z \otimes I + I \otimes Z$, the operator $X \otimes X + Y \otimes Y$ commutes with C, while $X \otimes I$ does not.*

Proof. The exchange operator preserves total Z charge; direct Pauli multiplication gives the stated commutators. □

This finite example demonstrates a constraint-invariant relational observable, not a solution of gravitational constraints. In the spatially compact vacuum setting, the local-observable obstruction excludes the relevant class of gauge-invariant observables built locally from finitely many metric derivatives [18]. It does not exclude relational, nonlocal, asymptotic, or matter-clock observables outside those hypotheses.

The classical hypersurface-deformation algebra contains, schematically,

$$\{H[N], H[M]\} = D[q^{ab}(N\partial_b M - M\partial_b N)], \tag{52}$$

together with the D–D and D–H brackets. The structure functions depend on the metric. A quantum representation must specify regulator, domains, operator ordering, boundary terms, physical states, and the continuum interpretation of closure. A small finite commutator residual is not enough.

Proposition 16.2 (Finite truncation can hide an algebra defect)**.** *For the oscillator truncated to N levels,*

$$[a_N, a_N^\dagger] = I - N|N-1\rangle\langle N-1|. \tag{53}$$

The defect vanishes on the lower-level block but has norm N on the full truncated space.

Proof. The diagonal entries of the commutator are one except at the highest level, where the missing raising transition gives $-(N-1)$. Subtracting the identity gives the formula. □

The example is not a gravitational anomaly. It shows why cutoff-boundary and continuum estimates are required before declaring an operator algebra represented. Gauge constraints and gravitational dressing also obstruct naive factorization of the physical Hilbert space into independent spatial factors. Microscopic tensor composition and physical constrained factorization are different questions.

17 Finite-resource inference and ultraviolet nonuniqueness

An observational obstruction must distinguish exact equality of laws from small differences and must specify the permitted experiments and resource budget.

Theorem 17.1 (Finite-resource discrimination bound)**.** *Suppose two models have single-trial laws P, Q with $\mathrm{TV}(P, Q) \le \varepsilon$. For N independent trials,*

$$\mathrm{TV}(P^{\otimes N}, Q^{\otimes N}) \le \min(1, N\varepsilon). \tag{54}$$

For equal prior probabilities, achieving average error at most $\delta < 1/2$ requires

$$N\varepsilon \ge 1 - 2\delta. \tag{55}$$

The same bound holds for an adaptive classical transcript if the conditional next-record laws differ by at most ε uniformly over all histories and allowed actions.

Proof. A maximal coupling fails to match a single record with probability at most ε. Couple trials sequentially and use the union bound, giving mismatch probability at most $N\varepsilon$. For adaptive records the same argument applies while histories agree, using the uniform conditional bound. Equal-prior optimal error is $(1 - \mathrm{TV})/2$, which yields (55). □

For quantum channel uses, an analogous hybrid argument uses the half-diamond distance, uniformly over permitted ancillas and inputs. A bound only for one fixed measurement is insufficient to constrain arbitrary coherent adaptive experiments.

Proposition 17.2 (A stronger regular statistical bound)**.** *If $D_{\mathrm{KL}}(P\|Q) \le \kappa$, then independent trials obey*

$$\mathrm{TV}(P^{\otimes N}, Q^{\otimes N}) \le \sqrt{N\kappa/2}. \tag{56}$$

Consequently error at most δ requires $N \ge 2(1 - 2\delta)^2/\kappa$. In regular nearby families with $\kappa = O(\varepsilon^2)$, the necessary scaling is ε^{-2} rather than the weaker ε^{-1} bound.

Proof. Relative entropy adds for independent products. Apply Pinsker's inequality and the equal-prior testing formula. For adaptive classical records the chain rule gives the same conclusion under a uniform conditional relative-entropy bound. □

Corollary 17.3 (Resource-relative resolution horizon)**.** *If every permitted protocol has a uniform per-use law difference bounded by ε and the total budget satisfies $N_{\max}\varepsilon < 1 - 2\delta$, no such protocol attains error δ. This is an approximate finite-resource obstruction, not exact operational equivalence.*

An EFT suppression $(E/M)^n$ is not automatically a total-variation bound. Converting it into one requires an observable, coefficient bounds, response functions, noise, state preparation, and control of all allowed protocols. Rare events, resonances, accumulated phases, and coherent strategies can change sensitivity. A cosmic-mode count is not a universal count of independent identical trials. Therefore no table of nominal energies proves that every ultraviolet completion lies in the exact kernel of all accessible observations.

Proposition 17.4 (Infrared data do not uniquely fix ultraviolet dynamics)**.** *If two models have the same retained low-energy coefficients and differ only in unmeasured higher-order terms, observations restricted to the retained order cannot select between them. Finite nonzero higher-order effects may nevertheless distinguish them with sufficient resources or a different protocol.*

Proof. At the retained order their predicted laws agree by construction. Beyond that truncation, equality has not been assumed, so no exact-kernel conclusion follows. □

Ultraviolet completion remains an open physical and mathematical problem. Neither a finite-resource bound nor a common Einstein limit proves that all candidate completions are consistent, equivalent, or permanently unobservable.

18 Further obstructions and their domains

Composite gravitons. The Weinberg–Witten result constrains massless spin greater than one carrying nonzero energy-momentum in a Lorentz-invariant setting with the relevant Lorentz-covariant conserved stress tensor [22]. A fixed-background composite-graviton proposal satisfying those hypotheses must address the obstruction. General relativity does not possess a local gauge-invariant gravitational stress tensor of the forbidden kind. Evading a hypothesis can remove the obstruction; it is not a construction of a consistent completion.

Classical horizons. For the fixed Schwarzschild geometry in ingoing Eddington–Finkelstein coordinates, outgoing radial null rays satisfy

$$\frac{\mathrm{d}r}{\mathrm{d}v} = \frac{1}{2}\left(1 - \frac{2M}{r}\right). \tag{57}$$

They do not escape from $r < 2M$ in that classical spacetime. This constrains direct causal access to interior events. It does not prove that all interior states have identical exterior quantum records: exterior fields, correlations, global constraints, and the quantum spacetime problem require separate analysis.

Reversibility and reduced dynamics. A reversible closed quantum evolution can induce irreversible reduced dynamics. A controlled-NOT acting on $|+\rangle|0\rangle$ produces an entangled pure state; discarding the second qubit leaves $I/2$. Thus Assumption A3 cannot be applied to every reduced subsystem to exclude decoherence. A measurement-and-feedback model may yield dephasing, but one such channel does not classify every classical–quantum hybrid. Decoherence–diffusion constraints are model-dependent physical tests [24], not a consequence of the polytope theorem alone.

No unique ultraviolet theory from exclusions. Eliminating polytopes under continuous transitivity, one matrix family under a composition rule, or an exact ghost pole does not exhaust all possible microscopic theories. Each exclusion is relative to a declared class. The derivation does not turn a finite list of no-go statements into an unrestricted uniqueness theorem.

19 A coherence-sensitive gravitational prediction

Assume the weak-field quantum Newton interaction

$$H_{\mathrm{int}} = -\frac{G m_A m_B}{|\hat{x}_A - \hat{x}_B|}. \tag{58}$$

For two equal-amplitude position branches per mass, negligible branch motion during the interaction, and controlled non-gravitational phases,

$$|\psi(t)\rangle = \frac{1}{2}\sum_{i,j=0}^{1} e^{i\varphi_{ij}}|ij\rangle, \qquad \varphi_{ij} = \frac{G m_A m_B t}{\hbar r_{ij}}. \tag{59}$$

Theorem 19.1 (Entangling phase invariant)**.** *The phase*

$$\Theta = \varphi_{00} + \varphi_{11} - \varphi_{01} - \varphi_{10} \tag{60}$$

is invariant under $\varphi_{ij} \mapsto \varphi_{ij} + \alpha_i + \beta_j$. The pure-state concurrence and negativity are

$$\mathcal{C} = \left|\sin\frac{\Theta}{2}\right|, \qquad \mathcal{N} = \frac{1}{2}\left|\sin\frac{\Theta}{2}\right|. \tag{61}$$

Proof. The local phases cancel in Θ. With coefficient matrix $A_{ij} = e^{i\varphi_{ij}}/2$, concurrence is $2|\det A| = |\sin(\Theta/2)|$. For a pure two-qubit state, the negative partial-transpose eigenvalue has magnitude equal to the product of Schmidt coefficients, which is half the concurrence. □

For $m_A = m_B = 10^{-14}\,\mathrm{kg}$, center separation $450\,\mu\mathrm{m}$, equal collinear branch splitting $250\,\mu\mathrm{m}$, and interaction time $2.5\,\mathrm{s}$, the stated geometry gives

$$\Theta = -0.313934492575, \qquad \mathcal{N} = 0.078161730662. \tag{62}$$

These are idealized calculated values in the assumed interaction model, with scales motivated by proposed entanglement experiments [23]. They are not observations. Finite wave-packet size, acceleration, trapping, environmental decoherence, electromagnetic interactions, and readout must be included in a physical error budget. Local dephasing can remove the ideal entanglement.

Proposition 19.2 (Classical local-channel obstruction)**.** *A separability-preserving local-operation and classical-communication channel cannot entangle an initially separable pair of probes. In particular, a convex mixture of product unitaries cannot do so.*

Proof. Each product operation maps a product state to a product state, or a conditional product branch. Convex mixing and classical conditioning preserve the convex set of separable states. Iterating the operations preserves separability. □

Inferring a nonclassical mediator from entanglement additionally requires initially uncorrelated probes, only the stipulated local probe–mediator couplings, no unaccounted quantum communication, and a classical mediator model restricted to the relevant classical-information channel class. The calculation does not identify a unique ultraviolet mediator, nor exclude every nonlocal direct-interaction or hybrid model without those assumptions.

20 The derivation theorem and dependency ladder

Theorem 20.1 (Conditional low-energy quantum gravity)**.** *Assume operational closure, the full quantum reconstruction conditions, an admissible quadratic causal continuum with calibrated volume (its signature and dimension then follow from Theorem 4.2 under its premises), a covariant source realization, the healthy universal tensor sector and its nonlinear consistency conditions, and perturbative quantum consistency. Then:*

(i) retained finite-dimensional records admit the complex quantum state/effect representation;

(ii) the causal realization and volume determine the physical Lorentzian metric within the admitted conformal class;

(iii) the source is the metric response, with its Ward identities, improvements, and renormalized correlations;

(iv) the leading two-derivative gravitational response is Einstein–Hilbert with universal coupling, and static tree exchange gives $-Gm_Am_B/r$;

(v) perturbative quantum predictions are organized by the gravitational EFT, not by Einstein-only closure;

(vi) physical predictions use gauge-invariant relational, dressed, asymptotic, or otherwise admissible observables;

(vii) the assumed coherent Newton interaction yields the invariant entangling phase of Theorem 19.1;

(viii) finite-resource discrimination is bounded by Theorem 17.1 and Proposition 17.2, without implying exact ultraviolet observational nullity;

(ix) the realization is $(1, 3)$*, one action quantum is universal, and thermal radiation obeys Theorem 15.3, whose* T^{n+1} *law, measured by absolute radiometry, independently witnesses* $n = 3$*;*

(x) a weight over histories that is a function of the Lorentzian action is an amplitude and not a probability, the free tensor kernel exists on the gauge quotient exactly for conserved sources, and it is not a probability kernel in either signature (Section 11);

(xi) in the scalar sector the interacting $3 + 1$ *limit is settled: the classical well-prepared limit exists at rate* a^2 *with predictor* $a^2 E_1$*, and the quantum limit obeys the interaction–resolution bound (32) (Section 9).*

Conclusions (x) and (xi) do not use the tensor-sector or perturbative assumptions; (xi) uses Structural input 9.10 for its quantum half.

Proof. Apply Structural input 2.4 to the reconstruction assumptions. Causal reconstruction and Theorem 4.4 fix the metric within the admitted realization. Theorem 7.1 identifies the source response. Theorem 10.1, Structural input 10.3, and Corollary 10.4 give the leading tensor dynamics. Proposition 14.1 fixes the static normalization. Assumption A8, Structural input 13.2, and Theorem 13.3 give the quantum low-energy organization. The observable construction must respect the constraints described in Section 16. Conclusions (vii) and (viii) are the separately proved phase and discrimination statements. Conclusion (ix) combines Theorem 4.2, Proposition 15.2, Theorem 15.3 and Corollary 15.4. Conclusion (x) is Theorem 11.3 with Propositions 11.4 and 11.5. Conclusion (xi) is Theorems 9.2 and 9.4 with Proposition 9.11. □

The finite-range construction supplies an explicit compatible localization and free-continuum example, and Section 9 an interacting scalar one. Neither is a premise-free derivation of the four-dimensional continuum or the metric-response assumption. Conversely, the main theorem does not require that particular walk to be the microscopic physical theory.

Step	Result	Dependence
1	Operational properties and predictive dynamics	Exact descent criterion; physical record family specified
2	Complex quantum representation	Full reconstruction assumptions and structural theorem
3	Coherence and phase invariants	Declared quantum realization and reference basis
4	Lorentzian signature, $D = 4$	Predictive closure; exchange and stable-carrier premises; Tensor Derivation
4b	Lorentzian metric	Admitted distinguishing continuum; causal reconstruction and calibrated volume
5	Exact finite propagation and free Dirac limit	Explicit one-dimensional finite-range unitary construction
6	Stress and quantum Ward identities	Covariant source functional and anomaly conditions
6b	Interacting scalar limit, predictor, resolution bound	Energy method with uniform L^6 bound; well-prepared data; triviality theorem imported
7	Einstein leading response	Linear selection proved; nonlinear consistency used structurally

Step	Result	Dependence
7b	Kernel existence, amplitude, conserved source	Quadratic action; weight a function of the action
8	Quantum gravitational EFT	Perturbative quantization, counterterms, matching, validity window
9	Observable phase and discrimination	Declared interaction and experimental/resource assumptions
9b	Action quantum, thermal spectrum	Exchange balance; Gibbs state; radiometric T^4 witness
10	Nonperturbative completion	Not implied by the preceding steps

21 Status ledger and remaining mathematical questions

The ledger records the precise scope and dependencies of the final statements.

Component	Status	Exact scope or missing input
Operational descent	PROVED	Factorization criterion; microscopic closure must be demonstrated
Quantum realization	CONDITIONAL	Reconstruction axioms not derived from finite resolution alone
Composition identity	PROVED	All ranks within the declared same-type matrix family
Coherence geometry	PROVED LOCALLY	Fixed reference/support; positivity and stratum changes retained
Signature obstruction	PROVED	Positive real pullbacks cannot be Lorentzian
Lorentzian signature	PROVED	Given a quadratic realization; predictive closure
Dimension $3+1$	CONDITIONAL	Exchange and stable-carrier premises; resolved scales only
Order and volume	CONDITIONAL	Unique metric within admitted causal continuum and calibrated measure
Strict propagation	CONSTRUCTED	Explicit free one-dimensional walk; exact support bound
Free continuum limit	PROVED	Fixed time and momentum band; error bound (14)
Quantum balance	PROVED	Number-current operator identity; not yet gravitational stress
Metric/source identification	CONDITIONAL	Covariant functional and record correspondence assumed; free level discharged (Proposition 11.5)
Kernel existence	PROVED	Quadratic actions; quotient, definite, and indefinite cases
Amplitude from signature	CONDITIONAL	Weight over histories assumed to be a function of the action
Gravitational kernel	IDENTIFIED	Signature $(5,1)$ on the gauge quotient; conformal-factor problem; linear order
Linear tensor response	PROVED	Declared local parity-even two-derivative ansatz
Nonlinear Einstein sector	CONDITIONAL	Structural consistency result with field-content hypotheses
Matter anomaly controls	CHECKED	Specified Standard-Model representation; not full quantum gravity

Component	Status	Exact scope or missing input
Perturbative EFT	CONDITIONAL	Gauge consistency, matching, and low-energy error control
Relational observables	REQUIRED	Full physical observable algebra not constructed here
Finite-resource inference	PROVED	Uniform law/channel bounds and explicit resource class
Gravitational entanglement	CONDITIONAL PREDICTION	Quantum Newton interaction and controlled protocol
Action quantum	ANCHOR, UNIVERSAL	Exchange balance; value measured
Thermal spectrum	PROVED	Gibbs state, Bose statistics; absolute radiometry witnesses $n = 3$
Classical interacting $3+1$ limit	PROVED IN CLASS	Quartic field, smooth well-prepared data, uniform lattice; rate a^2; endpoint of the method
Lattice-error predictor	PROVED	$a^2 E_1$; cubic-anisotropic dispersion; spacing bound is realization-specific
Isotropy and defect-free limit	MATCHING DATA	Not implied by refinement (Obstructions 9.6, 9.7)
Quantum scalar $3+1$ limit	CLOSED, NEGATIVE	Triviality imported for one component; bound (32) at one loop
Interacting gauge $3+1$ limit	OPEN	Asymptotically free sectors; stress limits; continuum existence
Quantum constraints	OPEN	Regulated closure, domains, physical Hilbert space
Ultraviolet completion	OPEN	Existence, unitarity, singular regimes, physical selection

Open problem 1 (Microscopic selection). Derive or empirically select the state-space, composition, and causal-realization assumptions from a concrete microscopic record/update law. A finite list of excluded alternatives is insufficient.

Open problem 2 (Interacting gauge continuum and stress convergence). Construct an interacting four-dimensional limit for asymptotically free gauge and fermion sectors, with uniform estimates controlling correlation functions, renormalized source variations, causal propagation, and any defect stress. The scalar sector is settled in Section 9: classically within the well-prepared class, and negatively at the quantum level. Two narrower questions remain there: the classical limit for data of finite energy only, and a proof of triviality for the four-component field.

Open problem 3 (Physical constraint representation). Construct a regulated quantum representation of the gravitational gauge/constraint identities with controlled domains and continuum limit, a physical state space, and a sufficient algebra of relational or dressed observables.

Open problem 4 (Ultraviolet existence and selection). Establish a consistent completion, including a definition of the gravitational amplitude kernel beyond quadratic order (Proposition 11.4), its controlled infrared matching, and predictions distinguishing it from other completions with the same retained EFT coefficients. Finite-resource limitations do not close this question.

22 Conclusion

The derivation identifies a precise low-energy endpoint: a quantum state/effect theory, a causally and volumetrically calibrated Lorentzian realization, a covariant stress response, a universal Einstein leading tensor sector, and its perturbative gravitational EFT. The construction includes

exact descent criteria, an all-dimension composition identity, finite propagation, an operator continuity law, a controlled free continuum bound, an interacting scalar limit with its error predictor, the existence conditions of the quadratic kernel, the linear tensor selection, and quantitative inference and entanglement formulas.

The remaining assumptions are physical and mathematical content, not merely notation. In particular, a local net is not yet a dynamical metric, internal phase curvature is not spacetime curvature, a finite resource horizon is not exact observational nullity, refinement does not by itself give an isotropic or defect-free limit, and a perturbative EFT is not a nonperturbative ultraviolet completion. Where the question could be decided it was: the quantum scalar field has no interacting limit of unlimited resolution, and the gravitational field has no probability kernel. The low-energy construction is consequently predictive within its stated window while leaving nonperturbative existence and microscopic selection as distinct mathematical questions.

A Computational verification and reproducibility

The accompanying calculations use constructed mathematical states and exact symbolic expressions; the one data-shaped check is synthetic and labelled as such, and three groups use published measured values (the radiometric constant, the Higgs mass, and a time-of-flight bound). A passing assertion witnesses an identity or example under the stated hypotheses; the interpretation of a theorem is determined by its hypotheses and proof.

The battery `rg_quantum_gravity.py`, with `rg_curv.py`, runs 30 groups in about twenty-five seconds with seed 20261110. Its output is `rg_quantum_gravity_results.json`, and reruns are byte-identical in the stated environment. Every group carries a canary.

Group	Content
QA–QC	polytope rigidity; composition factorization $-k(k-2)ab(a-1)(b-1)/4$; continuum canary
LA	positive-pullback no-go; Poisson volume calibration
LB, LD	exact cone and dispersion; free Dirac bound $TaP\|M\|$; $\omega'^2 \le 1$
LC, LE	odd anticommutator outside the cone; even bilinears commute outside, not inside
SA, SB	number-current operator balance; internal holonomy is not curvature
EA–ED	Newton normalization; Schwarzschild Riem^2, Riem^3; one-loop magnitudes; anomaly arithmetic
FA	conserved improvement; equal means with unequal fluctuations; truncation defect
CA	relational observable toy
UA–UC	TV and KL bounds (KL $\approx 2\varepsilon^2$), exact product TV; classical horizon; exact toy propagator
PA, PB	entangling phase and negativity, LOCC canary; reversible joint with irreversible reduced dynamics
HA, HB	universal action quantum; thermal spectrum in n dimensions, Wien roots, mode counting, synthetic shape test, radiometric witness, FIRAS circularity canary
KA	quadratic kernels: level-shell divergence, signature phase for five signatures, regulator independence, null-direction divergence, complex covariance
KB	linearized gravity: gauge zeros and one negative direction for $D = 3, \ldots, 6$; conserved-source existence; Lorentzian and Euclidean lattice forms; positive scalar Hessian
IA	uniform discrete L^6 bound: test families, adversarial ascent, walls in exponent and in dimension
IB	three-dimensional interacting refinement: state and two source variations at rate a^2, work balance, reference self-consistency, wrong-law canary

Group	Content
IC	predictor $a^2 E_1$ and error bound; cubic anisotropy; improved-stencil canary; spacing bound
ID	hidden alternating defect; laminate effective tensor and Bloch speed; two-limit sequence
IE	zero-point scaling; interaction–resolution bound; measured scalar coupling; asymptotic-freedom canary

Signature and dimension (Theorem 4.2) are checked in the Tensor Derivation battery, groups X1–X4. Numerical tests do not prove the imported reconstruction, nonlinear completion, two-loop calculation, triviality theorem, or quantum constraint closure. In groups IA and IB the numbers witness a proved estimate; they are not its proof. The script `rg_quantum_gravity_mutations.py` applies thirteen one-line falsifications to groups KA–IE, among them a wrong signature phase, a wrong linearization, an isotropic lattice correction, an arithmetic mean across the layers, and a scalar mass of 160 GeV; the battery reports each one as a failure of the targeted group.

B Changes in this edition

- Third edition: Section 11 (existence of the kernel) and Section 9 (the interacting $3+1$ limit) are new. The ledger row *Interacting* $3+1$ *limit: open* is replaced by five typed rows, and the corresponding open problem is narrowed to the gauge sectors. Seven battery groups are added.
- What the new sections do not claim: no derivation of the reconstruction conditions, no interacting quantum field in four dimensions, no statement about gauge sectors, and no gravitational content beyond the quadratic form.
- Signature and dimension are derived (Theorem 4.2) instead of being assumed in Assumptions A5 and A7.
- The action quantum, its universality, and thermal radiation with its dimension witness are added (Section 15).
- Correction to the first printing of this edition: the cosmic-spectrum dimension witness is withdrawn as circular and replaced by absolute radiometry (Corollary 15.4 and the remark after it).
- Several claims of the first print edition are withdrawn or narrowed, and the battery is rebuilt to match:
  - reversibility no longer excludes classical–quantum hybrids;
  - the resolution horizon is resource-relative;
  - Coleman–Mandula is not invoked;
  - the walk current is a number current, not a stress;
  - locality is graded.

C Normalization and approximation conventions

The spacetime signature is $(-,+,+,+)$ and $c = \hbar = 1$ except in dimensional estimates and the entanglement example. The reduced Planck mass satisfies $M_{\mathrm{Pl}}^{-2} = 8\pi G$ and $\kappa^2 = 32\pi G$. The mass parameter M in the free Dirac limit is distinct from a generic EFT cutoff; its meaning is local to Section 5. The Schwarzschild parameter is in geometrized units. The SLD/Bures normalization is $\mathrm{d}s_B^2 = F_Q \mathrm{d}\theta^2/4$. Total variation is half the L^1 distance, and relative entropy uses natural logarithms. Testing errors use equal priors unless stated otherwise.

The bibliography identifies the structural literature used in the derivation. Empirical scales are illustrative inputs rather than claims of the latest experimental limits, with three exceptions that are used as stated in their sources: the radiometric constant [25], the Higgs mass [37], and the quadratic time-of-flight bound [38]. No fit to experimental data is asserted.

References

[1] J. D. Bekenstein, Universal upper bound on the entropy-to-energy ratio for bounded systems, *Phys. Rev. D* 23 (1981) 287.

[2] Ll. Masanes, M. P. Müller, A derivation of quantum theory from physical requirements, *New J. Phys.* 13 (2011) 063001.

[3] L. Hardy, Quantum theory from five reasonable axioms, arXiv:quant-ph/0101012 (2001).

[4] G. Chiribella, G. M. D'Ariano, P. Perinotti, Informational derivation of quantum theory, *Phys. Rev. A* 84 (2011) 012311.

[5] M.-O. Renou et al., Quantum theory based on real numbers can be experimentally falsified, *Nature* 600 (2021) 625.

[6] S. W. Hawking, A. R. King, P. J. McCarthy, A new topology for curved space-time which incorporates the causal, differential, and conformal structures, *J. Math. Phys.* 17 (1976) 174.

[7] D. B. Malament, The class of continuous timelike curves determines the topology of spacetime, *J. Math. Phys.* 18 (1977) 1399.

[8] L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, Space-time as a causal set, *Phys. Rev. Lett.* 59 (1987) 521.

[9] R. Haag, D. Kastler, An algebraic approach to quantum field theory, *J. Math. Phys.* 5 (1964) 848.

[10] G. M. D'Ariano, P. Perinotti, Derivation of the Dirac equation from principles of information processing, *Phys. Rev. A* 90 (2014) 062106.

[11] A. Bisio, G. M. D'Ariano, P. Perinotti, Special relativity in a discrete quantum universe, *Phys. Rev. A* 94 (2016) 042120.

[12] A. A. Abdo et al., A limit on the variation of the speed of light arising from quantum gravity effects, *Nature* 462 (2009) 331.

[13] S. Coleman, J. Mandula, All possible symmetries of the S matrix, *Phys. Rev.* 159 (1967) 1251.

[14] G. 't Hooft, M. Veltman, One-loop divergencies in the theory of gravitation, *Ann. Inst. H. Poincaré A* 20 (1974) 69.

[15] M. H. Goroff, A. Sagnotti, The ultraviolet behavior of Einstein gravity, *Nucl. Phys. B* 266 (1986) 709.

[16] J. F. Donoghue, General relativity as an effective field theory: the leading quantum corrections, *Phys. Rev. D* 50 (1994) 3874.

[17] N. E. J. Bjerrum-Bohr, J. F. Donoghue, B. R. Holstein, Quantum gravitational corrections to the nonrelativistic scattering potential of two masses, *Phys. Rev. D* 67 (2003) 084033.

[18] C. G. Torre, Gravitational observables and local symmetries, *Phys. Rev. D* 48 (1993) R2373.

[19] S. A. Hojman, K. Kuchař, C. Teitelboim, Geometrodynamics regained, *Ann. Phys.* 96 (1976) 88.

[20] BICEP/Keck Collaboration, Improved constraints on primordial gravitational waves, *Phys. Rev. Lett.* 127 (2021) 151301.

[21] K. S. Stelle, Renormalization of higher-derivative quantum gravity, *Phys. Rev. D* 16 (1977) 953.

[22] S. Weinberg, E. Witten, Limits on massless particles, *Phys. Lett. B* 96 (1980) 59.

[23] S. Bose et al., Spin entanglement witness for quantum gravity, *Phys. Rev. Lett.* 119 (2017) 240401; C. Marletto, V. Vedral, *Phys. Rev. Lett.* 119 (2017) 240402.

[24] J. Oppenheim, C. Sparaciari, B. Šoda, Z. Weller-Davies, Gravitationally induced decoherence vs space-time diffusion: testing the quantum nature of gravity, *Nat. Commun.* 14 (2023) 7910.

[25] T. J. Quinn, J. E. Martin, A radiometric determination of the Stefan–Boltzmann constant and thermodynamic temperatures between −40 °C and +100 °C, *Phil. Trans. R. Soc. Lond. A* 316 (1985) 85.

[26] D. J. Fixsen et al., The cosmic microwave background spectrum from the full COBE FIRAS data set, *Astrophys. J.* 473 (1996) 576.

[27] G. W. Gibbons, S. W. Hawking, M. J. Perry, Path integrals and the indefiniteness of the gravitational action, *Nucl. Phys. B* 138 (1978) 141.

[28] K. Jörgens, Das Anfangswertproblem im Großen für eine Klasse nichtlinearer Wellengleichungen, *Math. Z.* 77 (1961) 295.

[29] K. Symanzik, Continuum limit and improved action in lattice theories (I): principles and φ^4 theory, *Nucl. Phys. B* 226 (1983) 187.

[30] L. Tartar, H-measures, a new approach for studying homogenisation, oscillations and concentration effects in partial differential equations, *Proc. Roy. Soc. Edinburgh A* 115 (1990) 193.

[31] P. Gérard, Microlocal defect measures, *Comm. Partial Differential Equations* 16 (1991) 1761.

[32] F. Murat, L. Tartar, H-convergence, in *Topics in the Mathematical Modelling of Composite Materials*, Birkhäuser (1997) 21.

[33] M. Aizenman, H. Duminil-Copin, Marginal triviality of the scaling limits of critical 4D Ising and ϕ^4_4 models, *Ann. of Math.* 194 (2021) 163.

[34] M. Lüscher, P. Weisz, Scaling laws and triviality bounds in the lattice ϕ^4 theory, *Nucl. Phys. B* 290 (1987) 25; 318 (1989) 705.

[35] N. Cabibbo, L. Maiani, G. Parisi, R. Petronzio, Bounds on the fermions and Higgs boson masses in grand unified theories, *Nucl. Phys. B* 158 (1979) 295.

[36] T. Hambye, K. Riesselmann, Matching conditions and Higgs mass upper bounds revisited, *Phys. Rev. D* 55 (1997) 7255, arXiv:hep-ph/9610272.

[37] S. Navas et al. (Particle Data Group), Review of particle physics, *Phys. Rev. D* 110 (2024) 030001, and 2025 update.

[38] Z. Cao et al. (LHAASO Collaboration), Stringent tests of Lorentz invariance violation from LHAASO observations of GRB 221009A, *Phys. Rev. Lett.* 133 (2024) 071501.

[39] M. Rada, *Resolution Geometry* (print edition) and companions: Tensor Derivation (second edition); Coherence Geometry; Cone Closure; Observation Limit; Observer Bounds, 2026.